Target-Based Communication under Credibility-Distorted Beliefs

Daniel Díaz

This Version: October 2026
Latest Version Available **here**

ABSTRACT: Angeletos and Sastry (2021) theoretically show that central bank (CB) target-based communication anchors inflation expectations better than its interest rate counterpart. However, more UK households report adjusting their expectations based on the interest rate, they under-react to new information, and experienced inflation lowers credibility among the cohorts that have lived through the strongest inflationary episodes. This raises the question of whether intensively investing in target-based communication would anchor expectations and deliver welfare improvements in this context. I explore it by proposing an expectation mechanism that unifies experience-based expectations with endogenous credibility, distorted by the diagnostic expectations of Bordalo et al. (2018), and incorporate it into a Representative Agent (RANK) and a Heterogeneous Agent (HANK) New Keynesian model. The main findings are: i) while in the RANK model the welfare loss is entirely due to distorted expectations, in the HANK model the burden shifts from the older cohorts, who remember differently, towards the younger cohorts, who spend differently; ii) a policy fully focused on communicating the target is unequivocally superior in reducing the loss for all cohorts; and iii) target communication and traditional monetary policy are complements in the RANK model and substitutes in the HANK model. Therefore, Target-based communication is a highly effective tool to anchor expectations when credibility is eroded, which CBs could implement more intensively with ample welfare gains.

1. Introduction

When inflation returned in 2021–2023, central banks, CB, henceforth, around the globe asked households to trust that it would come back to the target rapidly. In 2022, when inflation reached four-decade highs, central bankers spoke with one voice, confirming their commitment to return inflation to the target in an attempt to prevent a de-anchoring of expectations. For example, at the 2022 Jackson Hole symposium, the Federal Reserve Chair, Jerome Powell, stated that the "overarching focus right now is to bring inflation back down to our 2 percent goal" and that "we will keep at it until we are confident the job is done" (Powell 2022), a message he repeated a year later when he reminded the public that "two percent is and will remain our inflation target" (Powell 2023). Another example is the European Central Bank, which made the same request in November 2022, when its President, Christine Lagarde, argued that "the more confident the public is that inflation will return to 2% in a timely manner, the smoother the adjustment process will be" (Lagarde 2022), or the Bank of England, which in the same November told households that "it's our job to make sure that inflation returns to our 2% target" and that it expected "inflation to fall sharply from the middle of next year" (Bank of England 2022).

Communicating the inflation target has become a recurrent strategy of CB, with both theoretical and empirical support. On the one hand, Angeletos and Sastry (2021) theoretically show that communicating the inflation target anchors expectations better than communicating the policy instrument, while, on the other hand, Coibion et al. (2022) empirically find that communicating a simple statement about the 2% inflation target lowers inflation expectations as much as what the full policy statement does, and D'Acunto et al. (2020) find that target-based communication moves household expectations more than what an interest rate statement does as well.

However, for UK households these results contrast with what they themselves report when asked about the reasons behind the change in their inflation expectations. Available data until 2026 from the Bank of England Inflation Attitudes Survey show that, on average, 26% of them cite the level of interest rates, whereas only 16% cite the inflation target, and this gap is consistent across age, income, and education, among other characteristics. This raises the policy question of the effects of using target-centred communication to reach a larger share of individuals,

given its documented advantages in anchoring inflation expectations.

This paper analyse this question through the lens of distorted beliefs that incorporate how credibility is shaped by inflation experiences under diagnostic expectations, DE henceforth, two additional empirical regularities for UK households documented in this paper. The data show an under-reaction when new information is incorporated into inflation forecasts, in line with the consensus-level evidence of Bordalo et al. (2020) but with the opposite sign to the over-reaction that diagnostic expectations were designed to capture. Likewise, households that have experienced a higher inflation rate during their lifetime tend to display a lower level of credibility, in line with Salle et al. (2023), who find that remembered price surges lower trust in the CB.

To rationalize these empirical facts, I propose a mechanism that unifies experience-based expectations with endogenous credibility in how agents form the subjective distribution of inflation from which they then retrieve their point expectations. Intuitively, agents are uncertain about the anchor around which inflation gravitates, which is either the inflation they have experienced or the inflation target. This is precisely where CB communication is introduced, since it intentionally pulls the experienced anchor toward the inflation target. Through Bayesian updating, I define the subjective probability that the agent assigns to the uncertain anchor being the inflation target as subjective credibility, which is then endogenously incorporated into the agent's predictive distribution of inflation.

Intuitively, distorted beliefs determine different ex-ante real interest rates across age cohorts, which, for the same inflationary shock, cause different consumption responses between cohorts. On the one hand, older cohorts with a higher inflation experience and lower credibility tend to expect higher inflation and perceive a lower ex-ante real rate. This situation pushes them to non-optimally bring consumption forward relative to the mean, compared with the young cohorts, which perceive a higher ex-ante real rate, forcing them to non-optimally postpone consumption. The non-optimal consumption responses relative to the mean, the consumption wedges henceforth, derived from the different perceived rates naturally constitute the measure on which I analyze the distributional effects of the inflationary shock. By embedding the distorted beliefs mechanism into a general equilibrium framework such as a Representative Agent New Keynesian (RANK) model and a lifecycle Heterogeneous Agent New Keynesian (HANK) model, I next study the welfare and distributional effects of the intensity with which the

central bank communicates its target.

The findings are summarized as follows. First, for both models, the consumption wedges are strongly composed of the distortion of beliefs, with a substantial difference for the HANK model. While for the RANK model all wedges are attributable to the distortion of expectations, when wealth and income heterogeneity matter as in the HANK model, the Unhedged Interest Rate Exposure (URE) is of utmost importance for determining the distributional effects of monetary policy, as Auclert (2019) argues. This implies that while in the RANK model the loss is entirely due to differences in the perceived rate derived from distorted expectations, in the HANK model the welfare loss acquires a component of who spends differently. Therefore, the burden shifts from those who remember differently – the older cohorts whose inflation experiences diverge consistently from the target – towards those who spend differently – the younger cohorts with the highest Marginal Propensity to Consume (MPC) and the smallest asset positions. This result also changes the incidence of the cohorts' weights in the aggregate loss, since the young cohorts now have a larger share of the aggregate loss in the HANK model compared with the RANK model.

The second result echoes the findings of Angeletos and Sastry (2021) and Coibion et al. (2022), by showing that under this distorted beliefs framework, a policy fully focused on communicating the inflation target is unequivocally superior in reducing the welfare loss for all cohorts. Although communication cannot touch the DE component, by intensively communicating the target the CB incline beliefs towards the target and delivers the lowest loss within its reach.

The third result connects target communication with traditional monetary policy by analysing whether they are complements or substitutes within this framework. For the RANK model, intensively communicating the target delivers the lowest loss together with an aggressive interest rate response. This is intuitive since in the RANK model an aggressive policy rate response reduces inflation and shrinks the dispersion of expectations across cohorts, while communication removes the credibility term of every consumption wedge. Together, therefore, the two policies can be used as complements to deliver the lowest possible loss. However, for the HANK model, the results are the opposite. For a given level of loss, the central bank can choose between not communicating the target with a moderate rate reaction, or communicating the target intensively and reacting aggressively with the rate. In this case, the policies become substitutes. The reason

why is that in the HANK model a stronger rate response raises the realised real rate, which redistributes across cohorts through the exposure channel. Therefore, the central bank faces a trade-off between the two policies, since now the best policy is one that involves communicating the target with a passive interest rate response to limit the distributional channel that URE channel imposes.

This paper is connected with the the literature as follows. First, with the literate on measuring CB credibility. In his seminal article, Blinder (2000) settles on the definition that a CB is credible if people believe it will do what it says. There are, at least, two groups that have empirically tried to quantify CB credibility. The first, following a reduced-form approach, measures it from the distance between expectations and the CB inflation target, as in Bomfim and Rudebusch (2000) and Bordo and Siklos (2015), or by asking people directly, as in Haldane and McMahon (2018).

On the other hand, the second group makes credibility endogenous, dependent on past errors. For instance, Hommes and Lustenhouwer (2019) model it as the fraction of the population that trusts the target rather than extrapolating past inflation, consistent with McMahon and Rholes (2025), who show experimentally that people judge the central bank by its past forecast errors. In parallel, bringing in a long-run experiential component, Malmendier and Nagel (2016) show that households form inflation expectations from the inflation they have personally lived through, and Salle et al. (2023) find that remembered price surges lower trust in the CB. This paper connects both strands by modelling credibility as a belief that each cohort updates against its own inflation history, and by introducing it as a probability inside the agent's mind, which uses the same functional form as the population fraction of Hommes and Lustenhouwer (2019), delivering a predictive probability distribution with endogenous credibility.

The second literature branch that this paper relates studies how inflation expectations respond to news, particularly, whether they over- or under-weight them in their predictions. In an influential paper, Coibion and Gorodnichenko (2015) document that consensus forecasts under-react to news, and Bordalo et al. (2020) have shown that under-reaction at the consensus level may coexist with over-reaction at the individual level. Using the diagnostic expectations (DE) of Bordalo et al. (2018), it is possible to reconcile these over- and under-reaction differences. Simultaneously, L'Huillier et al. (2024) and Bianchi et al. (2024) have incorporated DE into the New Keynesian model, which helps to analyse general

equilibrium questions about this sort of distortion. This paper borrows the DE operator and applies it to the predictive distribution of inflation expanded by credibility, which allow me to distorting inflation expectations.

The third research area to which this paper connects is CB communication with the public. There is ample evidence that communication works well to anchor the expectations of the most informed agents in the economy, such as professional forecasters and financial markets. For example, Romer and Romer (2000) show that private forecasters update taking as reference the new information the Fed reveals through its policy actions, or Nakamura and Steinsson (2018) who show that policy announcements move professional forecasts and asset prices at high frequency. However, communicating with the general public has been a gigantic challenge for the CB and has often been unsuccessful, as analysed by Binder (2017) and Lamla and Vinogradov (2019), who find that policy announcements are largely unnoticed by households, and by Kumar et al. (2015), who show that even after decades of inflation targeting, firms' inflation expectations in New Zealand remain essentially unanchored. This lack of reach of the CB to the public limits the beneficial results of target communication documented by Angeletos and Sastry (2021), D'Acunto et al. (2020) and Coibion et al. (2022) discussed above. This paper takes that evidence and seeks to contribute to the discussion on how investing heavily in communicating the target can significantly improve household welfare.

Finally, this paper is also related to the literature on heterogeneous beliefs and redistribution in New Keynesian models. Auclert (2019) shows that the covariance between MPCs and URE determines the winners and losers of monetary policy. Guerreiro (2023) and Shabalina and Tzaawa-Krenzler (2024) independently study how imperfect information interacts with wealth heterogeneity by adding an additional covariance, between MPCs and beliefs, which come from disagreement or attention, respectively. This paper borrows the cohort belief structure of Guerreiro (2023) and the sequence-space representation of Auclert et al. (2021) to solve HANK models to analyse who gains, and how much, from CB communication when expectations are distorted in the way I propose.

The rest of this paper is organized as follows. Section 2 presents the stylized facts, Section 3 develops the inflation expectation formation process, Section 4 embeds it into the RANK and HANK models, Section 5 presents the results on target communication and welfare, and Section 6 concludes.

2. Empirical Motivation

This section presents the stylized facts about inflation expectations for the UK that motivate this paper. Throughout this section, a mix different data sources are used. I make intensive use of the Bank of England Inflation Attitudes Survey, the source for inflation expectations and for the preferred measure of central bank credibility. The cohort inflation histories are built from the Consumer Price Index published by the ONS, and the wealth measure comes from the ONS Wealth and Assets Survey. A detailed explanation of the variable construction and may be found in Appendix C, while subsection C.5.1 is reserved for descriptive statistics.

2.1. Stylized Facts

Fact 1. UK households under-react to inflation news.

"Household inflation expectations under-reaction to news."

The evidence comes from applying into our sample the forecast-error-on-revision test of Coibion and Gorodnichenko (2015), which evaluates whether expectation revisions predict the subsequent forecast errors. If this is true, then, the forecasts have not fully incorporated the news, which is evidence of a degree of information rigidity. For US professional forecasters in the period 1969–2014, they find a coefficient around 1.2 for the inflation rate, and the same positive sign for US consumers and firms, and for professional forecasters in twelve advanced economies, the UK among them.

In this spirit, Bordalo et al. (2020) apply an analogous test to an ample set of US series from the SPF and Blue Chip surveys over 1968–2016 and find that over- or under-reaction depends on the level of aggregation. On the one hand, at the aggregate level (consensus) there is under-reaction[1] as in Coibion and Gorodnichenko (2015), but at the individual level, forecasters over-react to their own revisions. They reconcile both facts with diagnostic expectations (Bordalo et al. 2018) under dispersed information, and estimate a diagnosticity parameter of about 0.5, which is in line with the values that L'Huillier et al. (2024) and Bianchi et al. (2024) independently estimate when incorporating Diagnostic Expectations (DE) into the New Keynesian model.

[1]Similar under-reaction to macroeconomic news, including inflation, is documented for Italian and German firms by Born et al. (2025).

I visually implement the test in Equation 1 using the BoE Inflation Attitudes Survey, with quarterly data from 2010Q1–2024Q4, with the caveat that its design is a repeated cross-section, thus, only the consensus version of the test may be performed:

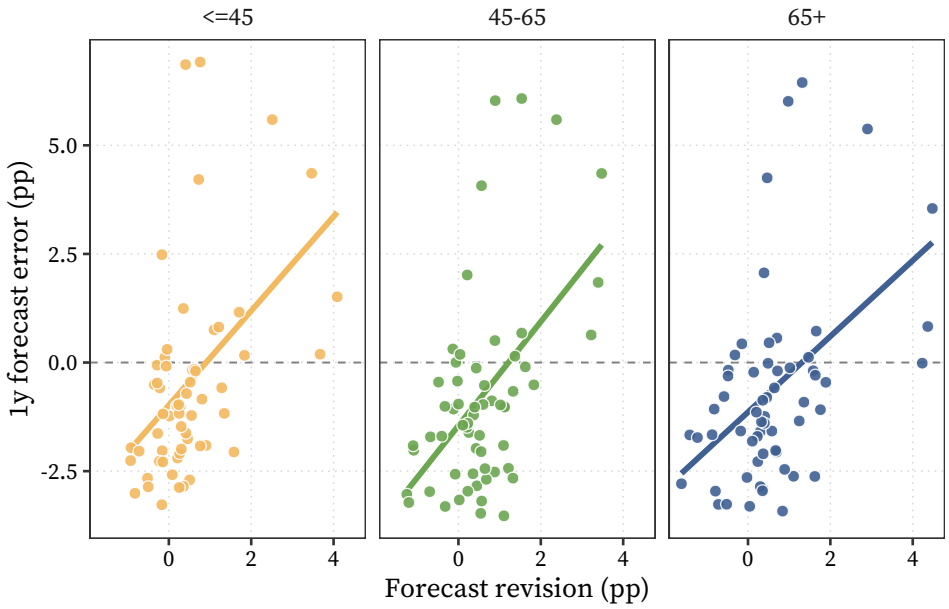
$$\underbrace{\pi_{t,t+3} - \mathbb{E}_{g,t}[\pi_{t,t+3}]}_{\mathrm{FE}_{g,t}} = \beta_{0,g} + \beta_{1,g} \underbrace{\left(\mathbb{E}_{g,t}[\pi_{t,t+3}] - \mathbb{E}_{g,t-4}[\pi_{t,t+3}]\right)}_{\mathrm{Rev}_{g,t}} + e_{g,t}. \quad (1)$$

Next, I provide a brief explanation of the elements of this regression. First, $\pi_{t,t+3}$ is the realised CPI inflation over quarters t to $t+3$, and is measured as the average of the four quarterly annualised rates. Second, $\mathbb{E}_{g,t}[\pi_{t,t+3}]$ is the cohort's consensus expectation over the same quarters, and $\mathbb{E}_{g,t-4}[\pi_{t,t+3}]$ is the expectation the same cohort held one year earlier for the same twelve months.[2]

In the spirit of Bordalo et al. (2022), revisions and errors are calculated on a fixed-event basis to keep a consistent comparison window. The expectation revisions, $\mathrm{Rev}_{g,t}$, show how expectations were updated today compared with one year ago for the same target window. On the other hand, forecast errors, $\mathrm{FE}_{g,t}$, show how close expectations were to realised inflation over the same window. Under-reaction is found when news that raise revisions are accompanied by positive forecast errors, which ultimately implies that the forecast fell short of fully incorporating the news.

[2]The different expectation's horizon, I use Q2a that asks "How much would you expect prices in the shops generally to change over the next 12 months?" and Q2b, that follows with "And how about prices the year after next? How much would you expect prices in the shops generally to change between [month, year +2] and [month, year +3]?".

FIGURE 1. **Forecast Errors and Revisions**



Note: Sample: BoE Inflation Attitudes Survey, quarterly waves 2010Q1–2024Q4, 60 waves per cohort.

Figure 1 shows a consistent positive slope regardless of age, meaning the three cohorts under-react by a similar amount. Thus, when a cohort revises its inflation expectation upward, realised inflation turns out to be above the revised forecast. Although the revision is made in the right direction, its magnitude is lower than what a full-information response would have implied, providing evidence that only a fraction of the news was incorporated in the revised forecast.

Fact 2. Lived Inflation, Credibility and Inflation Expectations.

"Cohorts that lived through higher inflation are less confident that inflation will return to the target, and cohorts that are less confident expect higher inflation."

This section documents that cohorts that have experienced higher inflation report lower credibility and also expect a higher inflation rate. To provide evidence on this fact, the credibility measure I use is question Q18 of the BoE Inflation Attitudes Survey with sample 2011Q1–2025Q3, when Q18 is asked. This question asks "How confident are you that inflation will be close to the 2% target set by the government in two to three years' time?"[3] I define the credibility of cohort g in

[3] By close it means between 1% and 3%. The potential answers are "very confident", defined in

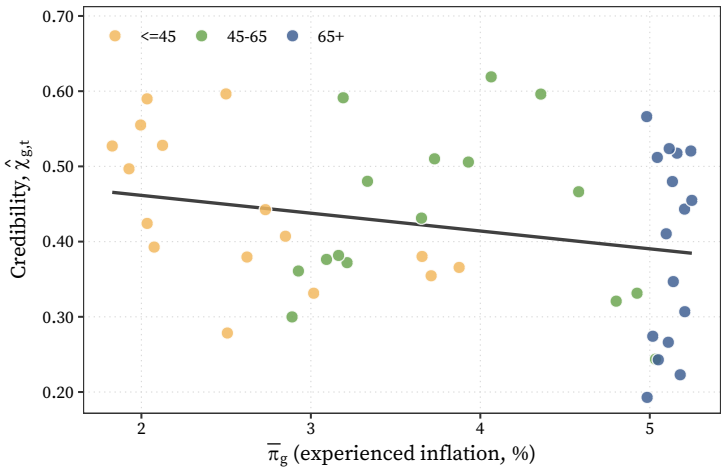
wave t as

$$\hat{\chi}_{g,t} \equiv \frac{\sum_{j\in(g,t)} w_j \mathbf{1}\{\text{Q18}_j \in \{\text{very confident, fairly confident}\}\}}{\sum_{j\in(g,t)} w_j}, \tag{2}$$

that is, the share of the cohort's respondents who answer "very" or "fairly" confident, among those the total. Therefore, we can consider it as the share of the cohort that trusts the target more than not. It is closely related to the credibility measure of Bomfim and Rudebusch (2000), in the sense of "the fraction of the population that believes the target will be achieved", and similarly to the fraction of fundamentalists of Hommes and Lustenhouwer (2019), who believe that inflation will be at the target.

Evidence for the first step of the fact is provided by Figure 2, which plots the cohort proportion self-reported credibility, $\hat{\chi}_{g,t}$, against the experienced inflation $\bar{\pi}_g$. It shows a negative relationship, which supports the intuitive scenario that more lived inflation is associated with lower credibility. This is consistent with the track-record view of Blinder (2000), in which credibility is earned by a history of low inflation, and with Bomfim and Rudebusch (2000), where the weight on the target falls with the past deviation of inflation from it and with Salle et al. (2023), who find that households who remember more price surges report lower trust in the central bank.

FIGURE 2. **Credibility and Experienced Inflation**

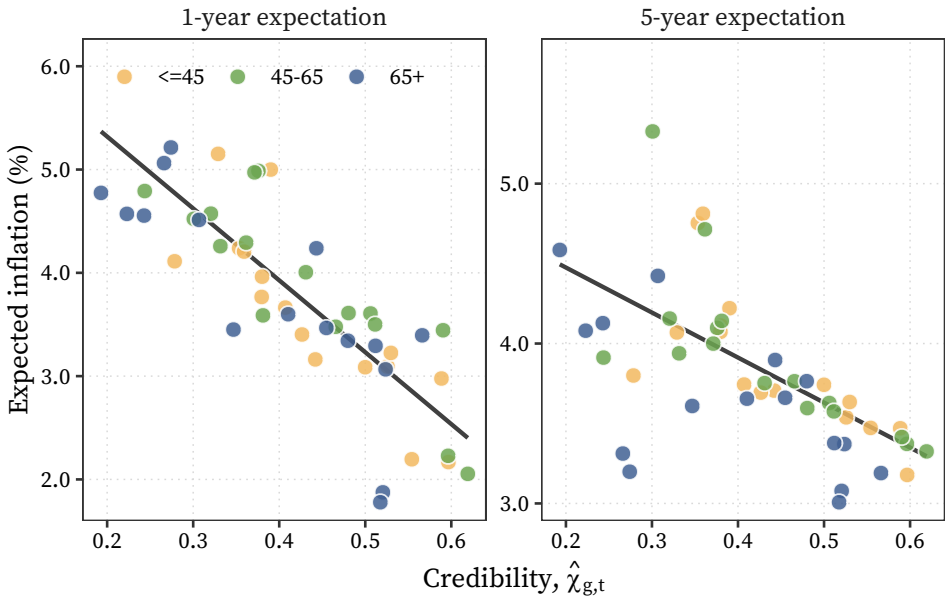


Note: Sample: BoE Inflation Attitudes Survey, waves with Q18, 2011Q1–2025Q3, 16 waves.

the questionnaire as a better than 50% chance, "fairly confident", around 50%, "not very confident", around 25%, and "not at all confident", less than 25%.

Figure 3 closes the reasoning by plotting the cohort's inflation expectations versus the self-reported credibility. At the two horizons, the slope is negative, meaning that a cohort with a lower proportion of people trusting that inflation will return to the target expects more inflation. This is consistent with Christelis et al. (2020), who find that households with more trust in the central bank report lower and less dispersed inflation expectations, and with Haldane and McMahon (2018), who show, using the same survey that respondents more satisfied with the Bank of England performance consistently get expectations closer to the inflation target. Conclusively, the two figures say that lived inflation lowers credibility and lower credibility raises expectations, which is the sequence that Section 3 builds into the proposed inflation expectations.

FIGURE 3. **Inflation Expectations and Credibility**



Note: Sample: BoE Inflation Attitudes Survey, waves with Q18, 2011Q1–2025Q3, 16 waves.

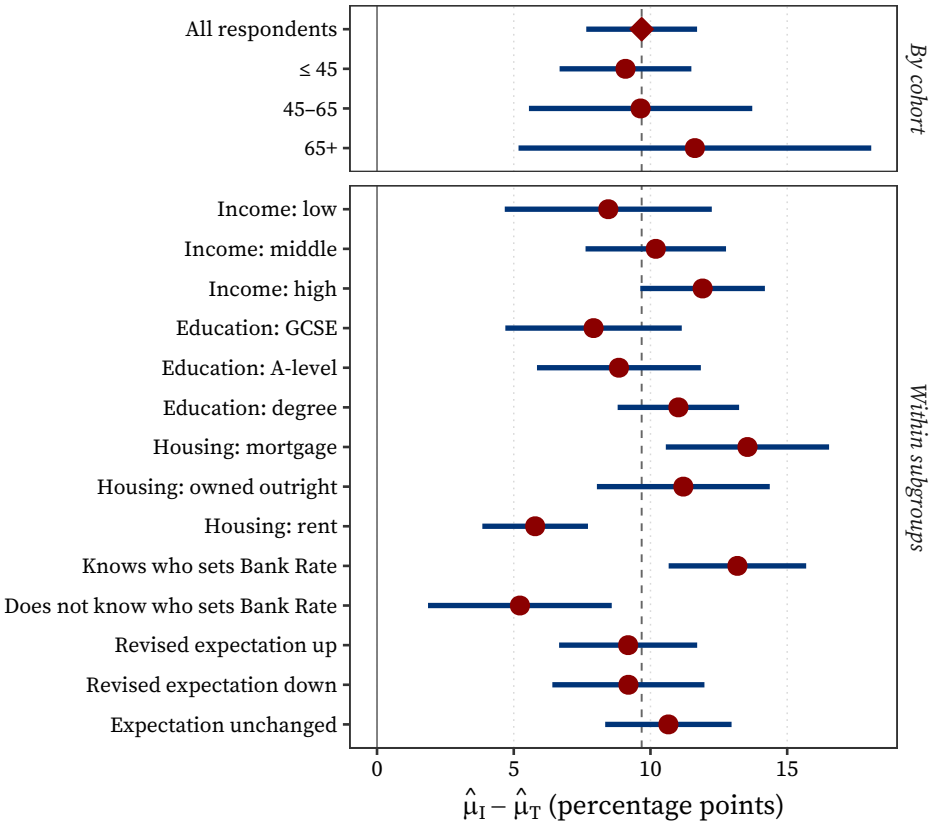
Fact 3. Instrument-Over-Target Salience.

"More households report forming their inflation forecast on the interest rate than on the inflation target."

To state this fact, I use question Q2aiv in the survey, which asks households who changed their expectation "What were the most important factors that led

you to your expectation of prices in the shops over the next 12 months?". Across the potential reasons, I focus on the cues the central bank controls through its communication: the inflation target, $\hat{\mu}_T$, and the policy instrument, $\hat{\mu}_I$. For each case, I report the mean across cohorts in Figure 4, which shows that the weighted share citing the instrument is statistically higher than the weighted share citing the target in every cohort and within subgroups of the population. This empirical result contrast with what Angeletos and Sastry (2021) theoretically recommend. In their framework, communication about the target anchors expectations more effectively than communication about the instrument when general-equilibrium feedbacks are strong. However, UK households report that they intensively form their inflation expectations around the policy rate.

FIGURE 4. **Instrument Salience: Interest Rate versus Inflation Target**



Note: Sample: BoE Inflation Attitudes Survey, waves with Q2aiv (2017Q1, 2023Q1, 2024Q1, 2025Q1, 2025Q3, 2026Q1). Income rows use the post-2022 sample. Dots are the estimates $\hat{\mu}_I - \hat{\mu}_T$, the vertical line marks the null $\mu_I - \mu_T = 0$, and horizontal lines are the 95% confidence intervals with clustered standard errors.

Three features of Figure 4 matter for the model presented in Section 3. First, the gap is positive and significant for every group, so a higher proportion of households report the interest rate as the cue they consider when changing their expectations. Second, the gap is indistinguishable across cohorts, so the salience of the instrument over the target is a common feature of the whole population rather than a characteristic of the cohorts that lived through high inflation. Finally, the difference is higher for high-income households and for those with exposure to monetary policy fluctuations, such as those with a mortgage and those who know who sets Bank Rate. This result is expected, since households with mortgages are exposed to the policy rate through their mortgage payments and savings, that is, through their balance sheet. This is the unhedged interest rate exposure channel documented by Auclert (2019).

This pattern is consistent with the evidence that central bank communication about its objectives is mainly ignored by the public. Binder (2017) and Lamla and Vinogradov (2019) find that households are mostly unaware of policy announcements, to the point that Haldane and McMahon (2018) describe the general public as the blind spot of central bank communication. Despite this unawareness of the public, there is evidence that the central bank can influence expectations with active target communication. Ehrmann et al. (2025) document that the announcement of the ECB's revised inflation target went largely unnoticed, whereas the same content delivered directly through information treatments raised the credibility of price stability. This is justifies why does the model of Section 3 considers the target as a feature the central bank can choose to actively communicate through the intensity ν.

Taken together, the three facts describe characteristic aspects of how UK households form inflation expectations. They under-react to news (Fact 1); the credibility they have is shaped by the inflation they have lived through, and inflation expectations are affected by it (Fact 2) and; the inflation target is cited less than the interest rate as a cue when changing their expectations (Fact 3). Section 3 takes these facts and formalizes a theory in which agents are uncertain about the anchor around which the inflation rate gravitates, credibility is microfounded within the agent's mind and, finally, under-reaction is an essential component explaining how agents incorporate news and ultimately delivers a distorted beliefs operator that will be incorporated in the upcoming sections.

Two additional facts, documented in Appendix D, show interesting evidence

on the potential effects that distorted beliefs may have on the economy. First, households that expect more inflation report bringing forward purchases and saving more (Fact A1). Second, cohorts that lived through higher inflation perceive higher inflation today and have become less confident that higher interest rates slow down price rises (Fact A2).

3. Credibility and Diagnostic Expectations

This section formally introduces how the credibility channel connects with the endogenous-credibility approach of Hommes and Lustenhouwer (2019) and the diagnostic-expectations (DE) framework of Bordalo et al. (2020) and L'Huillier et al. (2024). The connection is not direct, since DE can be used only when a well-defined subjective distribution is available. Particularly, DE are defined as a distortion of an agent's conditional distribution relative to that same agent's lagged conditional distribution. Therefore, to apply the diagnostic operator we need a forecaster whose beliefs at t and $t-1$ are linked by an updating rule. On the other hand, the endogenous credibility of Hommes and Lustenhouwer (2019) only provides the aggregate proportion of the population that is fundamentalist, its preferred measure of the anchored population and therefore of credibility. However, it leaves unanswered how agents incorporate credibility into their subjective distribution of future inflation at the individual level. Therefore, this section proposes an approach delivering a well-defined distribution for future inflation on which the diagnostic operator can act.

Suppose cohorts indexed by g, each with a representative agent within, as Assumption 1 states.

ASSUMPTION 1 (One forecaster per cohort). *Each cohort g is a single representative forecaster. Within a cohort there is no heterogeneity of beliefs and no averaging across people. Every weight that appears below is a probability held by this one agent, not a share of a population.*

This agent believes that the inflation rate is an AR(1) process around an anchor, m, whose true value is unknown. The perceived law of future inflation is summarized by Assumption 2:

ASSUMPTION 2 (Perceived Law of Future Inflation). *Cohort g believes inflation*

follows a first-order autoregression around an anchor m,

$$\pi_{t+1} = \rho_g \pi_t + (1 - \rho_g)\, m + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim N(0, \sigma_g^2)\ \textit{i.i.d.} \tag{3}$$

It knows ρ_g and σ_g, but it does not know m, the level inflation returns to.

Each agent balances two hypotheses about the anchor. First, under hypothesis T, the anchor is the central bank's inflation target. Second, under hypothesis E, the anchor is a communication-weighted sum of the inflation target and the lived inflation, $\bar{\pi}_g$. Introducing $\bar{\pi}_g$ as a potential anchor has ample empirical support. For instance, Malmendier and Nagel (2016) show that inflation expectations track the inflation they have lived through, as if each individual has estimated an AR(1) on its own history, assigning higher weight to recent observations. I assume under the latter, the central bank can reduce the influence of lived inflation on the anchor and displace it toward the inflation target. Assumption 3 formally states the two hypotheses.

ASSUMPTION 3 (Hypotheses About the Anchor). *Cohort g considers two values for the anchor,*

$$m_T = \pi^*, \qquad m_{E,g} = \nu\, \pi^* + (1 - \nu)\, \bar{\pi}_g, \qquad \nu \in [0, 1]. \tag{4}$$

Hypothesis T says the anchor is the target. Hypothesis E says the anchor is what the cohort lived through, moved toward the target by a fraction ν that measures the intensity of the central bank's target-communication. Finally, the lived inflation is given by $\bar{\pi}_g$.

To introduce central-bank communication into the formation of expectations, I focus on the content a central bank transmits when it talks about its inflation target. As documented by Binder (2017) and Lamla and Vinogradov (2019), policy announcements are generally unnoticed by the general public. However, an active target-communication campaign can succeed in shaping perceptions where passive announcement fails. For instance, Ehrmann et al. (2025) documented that the announcement of the ECB's 2021 monetary strategy review and the revised formulation of the inflation target went largely unnoticed by households. Despite household inattention, when the same content was delivered directly through randomized information treatments, the perceived credibility that price stability would be maintained increased significantly. Importantly, this paper does not provide an answer to the factors behind the apparent inattention or lack of interest

by the public when it comes to target communication. Instead, it assumes that the central bank can reach households if it intentionally decides to communicate its target intensively.

Accordingly, I assume the central bank directly chooses the *intensity* of its target-communication through $\nu \in [0, 1]$, where $\nu = 0$ is the baseline case in which the central bank is silent about the target and $\nu = 1$ is a target-communication strategy at its maximal intensity. Two properties of ν are worth mentioning. First, even when ν is a common communication strategy, its effects are cohort-specific, since it acts on each cohort's own experienced gap. Second, the intensity ν does not change the target or the lived inflation, but it changes what the cohort would believe under hypothesis E. Intuitively, a household that is consistently told what the inflation target is incorporates it into its alternative hypothesis, so its experiential anchor is pulled toward π^* by the fraction ν. Thus, under maximal target communication the two hypotheses coincide at the target, and credibility becomes irrelevant for the forecast.

Households do not know which hypothesis is right and learn about it from the forecast errors of each one of them:

$$e_{h,t} \;=\; \pi_t - \rho_g\, \pi_{t-1} - (1 - \rho_g)\, m_h. \tag{5}$$

Individuals incorporate hypothesis errors into future inflation beliefs via Bayesian updating. Hence, individuals calculate the log likelihood of each observation under each hypothesis given those errors as

$$\log f(\pi_t \mid \pi_{t-1}, h) \;=\; -\tfrac{1}{2} \log\!\left(2\pi\sigma_g^2\right) - \frac{e_{h,t}^2}{2\sigma_g^2}. \tag{6}$$

The learning occurs via discounting old evidence at a geometric rate ρ_U. This is the "forgetting" factor of Koop and Korobilis (2012) and essentially measures how much past performance is weighted relative to recent performance when evaluating hypothesis h. Besides, I explicity make the assumption that Assumption 4 formalise that idea.

ASSUMPTION 4 (Discounted Learning). *Cohort g starts with prior probability* 1/2 *on each hypothesis and updates by Bayes' rule with the discounted likelihood of each*

hypothesis $h \in \{T, E\}$*:*

$$\ell_{h,t} = \sum_{j \geq 0} \rho_U^j \log f(\pi_{t-j} \mid \pi_{t-j-1}, h), \qquad \rho_U \in (0,1). \tag{7}$$

Now, consider hypothesis T (the anchor is the inflation target) and calculate its posterior using Bayes' rule, and assume that ex-ante, before any error is accumulated, the prior the agent assigns to each hypothesis is 0.5, so there is no ex-ante bias across hypotheses. Also, following Koop and Korobilis (2012), assume that only the errors observed up to time $t-1$ are considered when forecasting at time t. I define the outcome of hypothesis T posterior updating as the *subjective credibility* and is summarized by Proposition 1.

PROPOSITION 1 (Subjective Credibility). *Under Assumptions 1–4, the probability that cohort* g *attaches to hypothesis* T *is*

$$\chi_{g,t} = \frac{e^{bU_{T,g,t-1}}}{e^{bU_{T,g,t-1}} + e^{bU_{E,g,t-1}}} = \left[1 + e^{-b\left(U_{T,g,t-1} - U_{E,g,t-1}\right)}\right]^{-1}, \qquad b = \frac{1}{2\sigma_g^2 (1 - \rho_U)}. \tag{8}$$

where the performance $U_{h,t}$ *of hypothesis* $h \in \{T, E\}$ *is*

$$U_{h,t} = -(1 - \rho_U)\, e_{h,t}^2 + \rho_U\, U_{h,t-1}. \tag{9}$$

PROOF. See Appendix A, subsection A.1. □

There are two immediate consequences of Proposition 1 that are closely related to the credibility proportion of Hommes and Lustenhouwer (2019). First, the posterior collapses to a logistic function of the difference in performance between hypotheses, which is why credibility takes the discrete-choice form as in their case. Second, the sensitivity parameter, b, contrary to their free "intensity of choice", here is the precision of inflation innovations corrected for forgetting. Hence, when inflation is noisy (high σ_g^2), a given forecast error is weak evidence against either hypothesis, and credibility moves slowly, but when inflation is predictable (low σ_g^2), the same forecast error moves credibility fast.

The next step is to calculate the predictive distribution of π_{t+1} weighted by credibility. I adopt the idea of averaging models that hold in different periods of time based on the "dynamic model averaging" of Koop and Korobilis (2012). Importantly, this approach incorporates credibility as part of the predictive distribution

of each agent by itself and not as a proportion of the population as in Hommes and Lustenhouwer (2019). Therefore, the predictive distribution from averaging both hypotheses can be calculated, and directly its associated expectation, as Lemma 1 states.

LEMMA 1 (Predictive distribution and the posterior-mean anchor). *Under Assumptions 1–4, cohort g's predictive distribution for π_{t+1} is the mixture*

$$\begin{aligned} f(\pi_{t+1} \mid I_t) &= \chi_{g,t}\, \varphi\big(\pi_{t+1}; \mu_{T,t}, \sigma_g^2\big) + (1-\chi_{g,t})\, \varphi\big(\pi_{t+1}; \mu_{E,t}, \sigma_g^2\big), \\ \mu_{h,t} &\equiv \rho_g \pi_t + (1-\rho_g) m_h, \end{aligned} \tag{10}$$

where $\varphi(x; \mu, \sigma^2)$ is the normal density, and its mean is

$$\begin{aligned} \mathbb{E}_{g,t}[\pi_{t+1}] &= \rho_g\, \pi_t + (1-\rho_g)\, m_{g,t}, \\ m_{g,t} &\equiv \chi_{g,t}\, \pi^* + (1-\chi_{g,t})\, m_{E,g} = \pi^* + (1-\chi_{g,t})(1-\nu)\, \Delta_g. \end{aligned} \tag{11}$$

where the distance between remembered inflation and the target,

$$\Delta_g \equiv \bar{\pi}_g - \pi^*,$$

is the cohort's experienced gap.

PROOF. See Appendix A, subsection A.2. □

As Equation (11) shows, the inflation expectation is the perceived law with the unknown anchor replaced by its posterior mean $m_{g,t}$, which is the target plus the part of the experienced gap that has not been removed by either credibility or communication, and which is ultimately where the central bank can exert its influence. Inflation expectations inherit the uncertainty about the anchor: when $\chi_{g,t} = 1$ (the *perfect-credibility scenario*), the anchor collapses to the target π^* regardless of the communication intensity selected by the central bank. At the opposite extreme, when $\chi_{g,t} = 0$ (the *null-credibility scenario*), the anchor collapses to $m_{E,g}$: expectations are anchored by the cohort's experiential inflation rate modulated by central-bank communication.

The ultimate step involves distorting the credibility-weighted predictive distribution by DE. Section 2 documents that household forecasts under-react to inflation news. Following Bordalo et al. (2018), I model this by assuming that the cohort does not report the mean of its predictive distribution but its *diagnostic*

version, which reweights each state by how much more likely it has become since last period. Two restrictions determine where the distortion acts. First, in line with the DE literature, Assumption 5 says that it acts on the forecast. I additionally assume that it takes credibility as a state that is already known, otherwise the operator would also distort average posterior. Thus, the same $\theta < 0$ that produces under-reaction in the forecast would make credibility under-react too, which would complicate interpretation since the credibility process would not be independently identified.

ASSUMPTION 5 (Diagnostic Distribution). *Cohort g does not report the mean of its predictive distribution. Instead, it reports its* diagnostic *version, the distribution*

$$f^{\theta}(\pi_{t+1} \mid I_t) \;\propto\; f(\pi_{t+1} \mid I_t)\left[\frac{f(\pi_{t+1} \mid I_t)}{f(\pi_{t+1} \mid I_{t-1})}\right]^{\theta}. \tag{12}$$

which reweights each value of π_{t+1} by how much more likely it has become since last period.

The parameter θ is common across cohorts. With $\theta > 0$ the cohort overweights states that became more likely (over-reaction), while $\theta < 0$ means it underweights them (under-reaction). Therefore, the distortion is applied to the predictive distribution of inflation under uncertainty about the anchor and takes credibility as a state of the perceived law that is already known when the forecast is made. That is, inside the agent's mind, she first sets her subjective credibility and then reports her diagnostic distribution. Additionally, Assumption 6 says that DE does not act on the learning process, which implies that hypotheses are evaluated by their own raw errors, and not by the errors of the distorted forecast.

ASSUMPTION 6 (Learning Independent of DE). *The two hypotheses, $\{T, E\}$, are compared by the errors of their own forecasts under Equation* (3)*, not by the error of the diagnostic forecast, so the learning probabilities are not distorted.*

Following L'Huillier et al. (2024), direct application of the resulting diagnostic distribution of Equation (12) leads to:

$$\mathbb{E}_t^{\theta}[\pi_{t+1}] \;=\; \mathbb{E}_t[\pi_{t+1}] + \theta\big(\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_{t+1}]\big). \tag{13}$$

Proposition 2 linearize this equation to ultimately deliver an inflation expectation that endogenously captures credibility and under-reaction to news, and the expression next sections will be adopted throughout.

PROPOSITION 2 (Diagnostic Expectation with Endogenous Credibility). *FOr cohort g, the Inflation Expectations linearized around the steady state, = π^*, is*

$$\mathbb{E}^{\theta}_{g,t}[\hat{\pi}_{t+1}] \;=\; \rho_g\,\hat{\pi}_t + (1-\rho_g)\,\kappa_g\,\hat{\chi}_{g,t} + \theta\,\rho_g\big(\hat{\pi}_t - \rho_g\,\hat{\pi}_{t-1} - (1-\rho_g)\,\kappa_g\,\hat{\chi}_{g,t-1}\big). \quad (14)$$

where κ_g is the credibility sensitivity of the anchor,

$$\kappa_g \;=\; -(1-\nu)\,\Delta_g. \quad (15)$$

PROOF. See Appendix A, subsection A.4. □

The persistence ρ_g captures the idea that a cohort that lived through persistent inflation expects any inflationary shock to last longer, and revises its expectations more. On the other hand, the credibility sensitivity κ_g shows that cohorts whose experienced inflation is close to the target respond only slightly to credibility changes, whereas cohorts whose inflation experience is far from the target exhibit larger expectation adjustments when credibility changes.

4. Credibility-Distorted Beliefs and the New-Keynesian Model

This section embeds the credibility-distorted beliefs of Section 3 into a Representative Agent New-Keynesian (RANK) model. The belief distortion essentially operates through the cohort's perceived real interest rate and the aggregate Euler equation, which ultimately causes cohorts to deviate from the FIRE intertemporal consumption path. Next, I extend the analysis to a richer model of the economy by considering a lifecycle Heterogeneous-Agent New-Keynesian (HANK) model in which cohorts also differ in their marginal propensities to consume (MPC) and in their balance-sheet exposure (unhedged interest-rate exposure, URE). This step makes clear how distorted beliefs now interact with the redistribution channel of monetary policy. While the *perceived real rate* affects consumption by distorting the intertemporal consumption path as in the RANK model, the *realised real rate*, the interest rate that households actually earn on their bonds, revalues each cohort's balance sheet through the URE channel. Hence, this distinction shows that the incidence in consumption shifts from the cohorts that forecast differently, as in the RANK model, to the cohorts that spend differently, as in the HANK model.

4.1. The Distorted RANK Model

This simple model consists of a discrete-time New Keynesian economy populated by a continuum of households divided into G cohorts. Every group is indexed by $g \in \{1, \ldots, G\}$ and has population weight $\omega_g \geq 0$, where $\sum_g \omega_g = 1$. Each cohort follows the same preferences over consumption $u(C) = C^{1-\sigma^{-1}}/(1-\sigma^{-1})$, derive disutility from labor $\zeta(N) = N^{1+\psi^{-1}}/(1+\psi^{-1})$, share the common income $y_{g,t} = y_t$, so there is no income heterogeneity. Households have access to a one-period nominal bond a_t paying the nominal interest rate i_t. Aggregate variables are expressed in uppercase letters while lowercase letters are reserved to deviations from the flexible-price steady state. Finally, the goods market clearing condition imposes $y_t = \int c_{i,t} \mathrm{d}i = \sum_g \omega_g c_{g,t}$, with $c_{g,t} \equiv \int_{i \in g} \frac{1}{\omega_g} c_{i,t} \mathrm{d}i$.

To cleanly isolate how the beliefs distorted channel shapes inflation expectations, I assume the beliefs about future output for cohort g are FIRE so, $\mathbb{E}_{g,t}[y_{t+h}] = y_{t+h}$ holds and anchored-recall only operates through inflation expectations and hence through the perceived real rate. While it may be possible that the distorted expectations developed affects how agents forecast future output, the empirical strategy used in Section 2 with the BoE Inflation Attitudes Survey does not have it counterpart for output expectations, so imposing a parallel channel on y would imply accepting a non-analyzed mechanism, which is out of the scope of this paper.

Each cohort g solves the maximization problem:

$$\max_{\{C_{g,t}, a_{g,t+1}, N_{g,t}\}_{t \geq 0}} \mathbb{E}^{\theta}_{g,0} \sum_{t=0}^{\infty} \beta_g^t \left\{ u(C_{g,t}) - \zeta(N_{g,t}) \right\} \tag{16}$$

subject to

$$C_{g,t} + a_{g,t+1} = Y_t + \left(1 + r_t\right) a_{g,t}, \tag{17}$$

and the no-Ponzi condition $\lim_{T \to \infty} \beta_g^T a_{g,T} \geq 0$ holds, which binds with equality at the optimum (transversality).

Importantly, Proposition 2 implies that for every horizon $h \geq 0$, cohort g's perceived ex-ante real rate is

$$r^{\text{per}}_{g,t+h} = i_{t+h} - \rho_g \pi_{t+h} - (1-\rho_g) m_{g,t+h} - \theta \rho_g \left[u_{t+h} - (1-\rho_g) m_{g,t+h-1} \right]. \tag{18}$$

In this simple version of the economy, absent of wealth accumulation, the

realised real rate, r_t, only affects the return of current assets and not intertemporal consumption decisions, which depend on the perceived rate $r^{per}_{g,t}$ alone. Moreover, since cohorts hold no assets in steady state, $a^{ss}_g = 0$, there is no stock of wealth to be revalued when r_t moves, so the realised rate has no effect on consumption. This is the key difference with the HANK model, where there are heterogeneous asset positions, so the same movement in r_t revalues their cash-on-hand, the URE channel.

Intertemporal consumption decisions are pinned down by the perceived real rate of cohort g. This rate depicts the cohort-specific channels through which the distortion enters into the consumption-saving decision. The first mechanism states that cohorts that lived through persistent inflation (high ρ_g) perceive a lower real rate, whereas low-persistence cohorts perceive a real rate closer to the nominal rate i_{t+h}. The second channel propagates central-bank credibility $\chi_{g,t+h}$ through $\kappa_g = \pi^* - m_{E,g}$. It asserts that cohorts whose experienced inflation has persistently diverged from the target react more strongly to credibility changes. Finally, the diagnostic parameter θ measures how the cohort reacts to news. When $\theta < 0$, the cohort under-reacts to news, so the perceived real rate falls by less than what the inflation news would demand, therefore households bring consumption forward by a lower magnitude of what the FIRE case would justify.

The standard solution of the household maximization problem leads to the cohort-g distorted Euler equation:

$$C^{-1/\sigma}_{g,t} \;=\; \beta_g(1 + r^{\text{per}}_{g,t})\,\mathbb{E}^{\theta}_{g,t}[C^{-1/\sigma}_{g,t+1}] \tag{19}$$

It is noteworthy mentioning the degree of cohort heterogeneity introduced by β_g, permitting the differences across cohorts in the perceived real interest rate will prevail in steady-state, $r^{per\,ss}_g = \beta_g^{-1} - 1$. The linearized version around the steady state for the perceived rate obeys

$$\hat{r}^{\text{per}}_{g,t+h} \;=\; \hat{\imath}_{t+h} - \rho_g\,\hat{\pi}_{t+h} - (1-\rho_g)\,\kappa_g\,\hat{\chi}_{g,t+h} - \theta\,\rho_g\big(\hat{\pi}_{t+h} - \rho_g\,\hat{\pi}_{t+h-1} - (1-\rho_g)\,\kappa_g\,\hat{\chi}_{g,t+h-1}\big). \tag{20}$$

while, Lemma 2 shows the equivalent expression for consumption:

LEMMA 2 (The cohort-g log-linearized consumption). *The first-order log-linearised*

solution of (16)–(17) *around* $a_g^{ss} = 0$, $\pi^{ss} = \pi^*$, χ_g^{ss}, *and* $r_g^{per\,ss} = \beta_g^{-1} - 1$ *is*

$$c_{g,t} \;=\; (1-\beta_g)\sum_{h=0}^{\infty}\beta_g^{h}\,y_{t+h} \;-\; \sigma\sum_{h=0}^{\infty}\beta_g^{h+1}\,\hat{r}^{per}_{g,t+h}, \tag{21}$$

where $\hat{r}^{per}_{g,t+h}$ *is given by* (20).

PROOF. See Appendix A, subsection A.5. □

As a final step, assume that all cohorts smooth consumption at roughly the same speed, which implies $\beta_g^h \approx \bar{\beta}^h$ for all g, h, and define $\bar{\beta} \equiv \sum_g \omega_g \beta_g$, the population-average perceived rate $\bar{r}_t^{\text{per}} \equiv \sum_g \omega_g \hat{r}_{g,t}^{\text{per}}$, and recall the goods-market clearing condition $y_t = \sum_g \omega_g c_{g,t}$. Then, Equation 21 may be simplified to:

$$y_t \;=\; (1-\bar{\beta})\sum_{h=0}^{\infty}\bar{\beta}^{h}\,y_{t+h} \;-\; \sigma\sum_{h=0}^{\infty}\bar{\beta}^{h+1}\,\bar{r}^{\text{per}}_{t+h}. \tag{22}$$

Cancelling common terms ultimately leads to:

$$y_t \;=\; y_{t+1} \;-\; \sigma\,\bar{r}_t^{\text{per}}.$$

Thus, the aggregation of the cohort-g Euler equations reduces to the representative-agent Euler equation.

To close the model, firms set prices à la Calvo and have perfect-foresight which yields the standard New Keynesian Phillips curve, augmented with an exogenous cost-push shock ε_t^{CP} that follows an AR(1) process with persistence ρ. The central bank conducts monetary policy through a Taylor rule that responds to inflation and is subject to an exogenous shock ε_t. Together with the aggregate Euler equation derived previously, the equilibrium of this simple economy is characterized by the following three-equation block:

$$\text{Policy Rule:}\;\; i_t = \phi\pi_t + \varepsilon_t, \tag{23}$$

$$\text{NKPC:}\;\; \pi_t = \varpi\, y_t + \beta\pi_{t+1} + \varepsilon_t^{CP}, \tag{24}$$

$$\text{Aggregate Euler:}\;\; y_t = y_{t+1} - \sigma\,\bar{r}_t^{\text{per}}, \qquad \bar{r}_t^{\text{per}} \equiv \sum_g \omega_g\,\hat{r}_{g,t}^{\text{per}}. \tag{25}$$

The three-equation block (23)–(25), together with the distorted expectations of Proposition 2, is the system that the RANK simulations solve. Importantly for the welfare analysis developed later in the paper, cohorts react more or less strongly

than the average, depending on their own distortion parameters ρ_g and κ_g, through their perceived real interest rate. To bring the model closer to the HANK literature, the next step is to introduce uninsurable income risk and heterogeneous asset positions, so that cohorts would differ in their marginal propensities to consume (MPCs) and in the exposure (URE) of their balance sheets to r_t.

4.2. The Distorted HANK Model

This model builds on the lifecycle HANK of Auclert et al. (2024), augmented with the distorted expectations mechanism introduced in Section 3 and the cohort belief heterogeneity structure of Guerreiro (2023). The economy consists of a continuum of infinitely-lived households $i \in [0,1]$, where each household belongs to an age-cohort $g \in \{1,\dots,G\}$. Every cohort has a population weight $\omega_g \geq 0$, $\sum_g \omega_g = 1$ and a discount factor $\beta_g \in (0,1)$. As standard in the HANK literature, I assume that households face uninsurable idiosyncratic labour-productivity risk $e_{i,t}$, which follows a Markov chain with transition matrix $P(e,e')$. This matrix is common across cohorts and has stationary distribution $\pi^e(e)$ such that $\sum_e \pi^e(e)\, e = 1$.

Every household in group g receives a post-tax labour income from supplying $N_{g,t}$ hours of labor (determined by unions) given by:

$$y_{g,t}(e) \;=\; (1-\tau_t)\, e\, w_t\, N_{g,t}, \tag{26}$$

where w_t is the real wage and τ_t the labour-income tax. Finally, markets are incomplete and households trade a one-period real bond a such that $a \geq 0$.

To keep consistency with Proposition 2, I introduce the diagnostic-expectations mechanism *only* through distorted beliefs about the aggregate inflation path, which in turn shape the perceived real interest rate that drives consumption decisions through the *Fisher channel*. By contrast, I assume that households observe their idiosyncratic state $e_{i,t}$ and are fully rational with respect to it. Moreover, they know the transition matrix $P(e,e')$ and form correct expectations about their own income. Accordingly, the cohort-g perceived ex-ante real rate is,

$$r^{\text{per}}_{g,t} \;\equiv\; i_t - \mathbb{E}^{\theta}_{g,t}[\pi_{t+1}]. \tag{27}$$

Importantly, distorted beliefs operate only at the cohort level. The real return that wealth actually earns is the *realised* ex-post real rate, $r_{t+1} = i_t - \pi_{t+1}$. Thus, a bond's realised return does not depend on what its holder believes, so every

household earns r_{t+1} on its bonds *regardless of its expectations*. The cohort-perceived rate $r^{\text{per}}_{g,t}$ in (27) is the belief about that same return held at the decision date, namely the Fisher equation evaluated under the diagnostic expectation $\mathbb{E}^{\theta}_{g,t}$, and it is the rate that *shapes inter-temporal consumption decisions*. The two rates therefore coexist as a realised return and a forecast of it. This distinction does not emerge in the RANK model, where households hold no wealth in steady state, so the realised rate has nothing to revalue and only $r^{\text{per}}_{g,t}$ enters the Euler equation, as discussed before.

For every cohort-g household entering period t with bond holdings a and productivity e, the consumption and savings decisions are obtained from solving the Bellman equation

$$\begin{aligned} V_{g,t}(a,e) \;=\; \max_{c,\, a' \geq 0} \; & u(c) \;+\; \beta_g \, \mathbb{E}^{\theta}_{g,t}\big[\, V_{g,t+1}(a',e') \,\big|\, e \,\big] \\ \text{s.t.} \quad & c + a' \;=\; y_{g,t}(e) \;+\; \big(1 + r_t\big)\, a, \end{aligned} \tag{28}$$

where $u(c) = c^{1-\sigma^{-1}}/(1-\sigma^{-1})$. The solution to Problem (28) defines the cohort consumption and asset policy functions $c_{g,t}(a,e)$ and $a_{g,t}(a,e)$.

Remarkably, the budget constraint in (28) uses r_t on existing wealth, while the optimal saving decision discounts future marginal utility at the cohort-perceived rate. This implies the distorted Euler equation:

$$u'(c_{g,t}) \;=\; \beta_g \big(1 + r^{\text{per}}_{g,t}\big)\, \mathbb{E}^{\theta}_{g,t}\big[\, u'(c_{g,t+1}) \,\big], \tag{29}$$

so the discount rate between t and $t+1$ uses the nominal rate and the diagnostic expectation of the decision date t, exactly as in the model of Section 4.1.

As discussed in Guerreiro (2023), Problem (28) imposes the law of iterated expectations at the individual level, $\mathbb{E}^{\theta}_{g,t}[\mathbb{E}^{\theta}_{g,t+1}[\cdot]] = \mathbb{E}^{\theta}_{g,t}[\cdot]$, so households behave as if they do not expect to revise their inflation beliefs. Since diagnostic expectations do not distort the idiosyncratic state, the cross-sectional distribution of beliefs is orthogonal to (a,e). This condition allows the behavioural Jacobians of Section 4.2.2 to be computed from their FIRE counterparts.

Finally, denote by $D_{g,t}(a,e)$ the joint distribution of assets and productivity.

Given the policy $a_{g,t}(\cdot)$ and the exogenous chain P, it evolves according to[4]

$$D_{g,t+1}(a', e') \;=\; \sum_{e} \int \mathbf{1}\{a_{g,t}(a, e) = a'\}\, P(e, e')\, D_{g,t}(\mathrm{d}a, e). \tag{30}$$

Hence, cohort aggregates may be calculated as

$$C_{g,t} \;=\; \sum_{e} \int c_{g,t}(a, e)\, D_{g,t}(\mathrm{d}a, e), \qquad A_{g,t} \;=\; \sum_{e} \int a_{g,t}(a, e)\, D_{g,t}(\mathrm{d}a, e), \tag{31}$$

while aggregate consumption and bond demand are given by $C_t = \sum_g \omega_g C_{g,t}$ and $A_t = \sum_g \omega_g A_{g,t}$.

The rest of the model is standard. The supply side of the economy, common across cohorts, follows Guerreiro (2023) and is the canonical sticky-wage New Keynesian block of Auclert et al. (2024). A competitive final-good firm operates the linear technology $Y_t = \Theta_t N_t$, so the real wage equals marginal productivity, $w_t = \Theta_t$. Price inflation is therefore wage inflation net of productivity growth. As in Guerreiro (2023), a continuum of unions aggregates cohort labour into N_t, sets wages subject to a quadratic adjustment cost, and delivers the linearised wage Phillips curve

$$\pi^{w}_{t} \;=\; \kappa_w \Big[\sigma^{-1} \hat{C}_t + \psi^{-1} \hat{N}_t - \big(\hat{w}_t - \hat{\tau}_t\big) \Big] \;+\; \bar{\beta}\, \mathbb{E}_t[\pi^{w}_{t+1}] \;+\; \varepsilon^{CP}_t, \tag{32}$$

with κ_w the slope, ψ the Frisch elasticity, ε^{CP}_t an exogenous cost-push shock, and hats denoting log-deviations. Cohort hours are allocated by the uniform union rule $N_{g,t} = \Gamma_g(N_t)$, with $\sum_g \omega_g \Gamma_g(N_t) = N_t$. Unions set wages and ration hours uniformly across households, so households do not choose labour directly and N does not enter their intertemporal problem. The government issues real debt $B_t = B$, spends $G_t = G$, and adjusts τ_t to satisfy $G + (1 + r_t)B = \tau_t Y_t + B$, and monetary policy follows a Taylor rule with an exogenous monetary shock ε^{MP}_t,

$$i_t = r_{ss} + \phi_\pi \pi_t + \varepsilon^{MP}_t, \qquad r_{t+1} = i_t - \pi_{t+1}. \tag{33}$$

Finally, the equilibrium for this economy is defined in Appendix B.

[4] I implement the lottery method described in Young (2010) on a grid, so (30) becomes $D_{g,t+1} = \Lambda'_{g,t} D_{g,t}$ for a sparse transition matrix $\Lambda_{g,t}$.

4.2.1. Sequence-Space Representation

This paper uses the sequence space approach introduced by Auclert et al. (2021) due to its effectiveness for solving heterogeneous-agent models when nonlinearities do not arise. In order to obtain the first-order transition dynamics, the economy is written as a set of blocks, each being fed by inputs to produce time paths of aggregate outputs. Define the first-order behaviour of a generic block with output O and input X as its Jacobian matrix

$$\mathcal{J}^{O,X} = \left[\partial O_t / \partial X_s\right]_{t,s\geq 0}, \tag{34}$$

whose (t, s) entry captures how the date-t output responds to a unit innovation in the date-s input, evaluated at the steady state. Intuitively, column s will be interpreted as the impulse response of O to news, received today, about the date-s value of X. Following Auclert et al. (2021), the household blocks' Jacobians are computed with the fake-news algorithm.

The economy described previously contains G cohort household blocks plus the production, fiscal, bond-pricing, Phillips-curve, monetary-policy, diagnostic-expectations, credibility, and market-clearing blocks. Any block other than the g household problems is a simple analytical block that resembles Galí (2015). Each cohort g block is a standard incomplete-markets recursion, and these are the only non-trivial blocks.[5]

Through the functional $\mathcal{C}_g(\cdot)$, cohort g takes as inputs the realised real rate r_t, the perceived real rate $r_{g,t}^{\text{per}}$ of (27), and its after-tax labour income $(1-\tau_t)\, w_t N_{g,t}$, and returns as outputs cohort consumption $C_{g,t}$ and asset holdings $A_{g,t}$ with

$$\mathcal{C}_g\big(\{r_t, r_{g,t}^{\text{per}}, Y_{g,t}, \tau_t\}_{t\geq 0}\big) = \sum_e \int c_{g,t}(a,e)\, D_{g,t}(\mathrm{d}a, e), \tag{35}$$

and analogously for asset demand $\mathcal{A}_g$. The same block returns each cohort's consumption, with $\sum_g \omega_g \mathcal{C}_g = C$, used to calculate the cohort-level wedges for the

[5] I adopt the same solution framework as in Auclert et al. (2021). Given the paths of inputs for every heterogeneous g block, I solve the problem (28) backwards for the marginal value of wealth and the consumption and savings policies $c_{g,t}(a,e)$, $a_{g,t}(a,e)$ using the endogenous-gridpoint method, over a state of idiosyncratic productivity e (evolving on the Markov chain $P(e,e')$) and bond holdings a on a grid. Then, I iterate the cohort distribution $D_{g,t}(\mathrm{d}a, e)$ forward and aggregate the policies. Cohorts follow the same recursion and the only differences are the discount factor β_g, the labour income $y_{g,t}$, and the associated perceived rate $r_{g,t}^{\text{per}}$. Finally, time paths are truncated at a horizon T, and the household Jacobians are computed once at the steady state.

HANK model in the next section.

The first-order response of cohort consumption $\mathbf{C}_g$ is a linear combination of its inputs' Jacobian contributions[6]

$$\mathrm{d}\mathbf{C}_g = \mathcal{J}_g^{C,r} \cdot \mathrm{d}\mathbf{r} + \mathcal{J}_g^{C,r^{\mathrm{per}}} \cdot \mathrm{d}\mathbf{r}_g^{\mathrm{per}} + \mathcal{J}_g^{C,Y} \cdot \mathrm{d}\mathbf{Y}_g + \mathcal{J}_g^{C,\tau} \cdot \mathrm{d}\boldsymbol{\tau}. \tag{36}$$

The Jacobian $\mathcal{J}_g^{C,r}$ is the income (cash-on-hand) consumption response to the realised real rate. This object matters because it captures the pure redistribution channel of the realised interest rate, whose impact response across cohorts reproduces the unhedged interest-rate exposure (URE) of Auclert (2019). Intuitively, the URE measures each household's net maturing position and its refinancing needs every period:

$$\text{URE} = \text{income} + \text{maturing assets} - \text{consumption} - \text{maturing debt}.$$

A household with $\text{URE}_g > 0$ is a short-term net saver, namely, a rise in the real rate raises its cash-on-hand, since it reinvests maturing positions at the higher rate, and it increases consumption. Conversely, a household with $\text{URE}_g < 0$ is a short-term net borrower since the same rise lowers its cash-on-hand, as it must refinance at the higher rate, and reduces consumption. Auclert (2019) argues that URE is a sufficient statistic to determine the winners and losers of any interest rate change since $\mathrm{d}C^r = \sum_g \omega_g \,\mathrm{MPC}_g \,\mathrm{URE}_g \,\mathrm{d}r$. Hence, only the joint distribution of marginal propensities and exposures matters to determine this consumption response. Since young cohorts, which hold the smallest asset positions, have the highest MPC, this covariance is negative and the redistribution channel amplifies monetary transmission in their model.

Distorted beliefs introduce an additional mechanism through which the central bank's decisions may trigger redistribution. Expanding $\mathcal{J}_g^{C,r^{\mathrm{per}}}$ in Equation 36 gives:

$$\begin{aligned}\mathcal{J}_g^{C,r^{\mathrm{per}}} \mathrm{d}\mathbf{r}_g^{\mathrm{per}} = {} & \underbrace{\mathcal{J}_g^{C,r^{\mathrm{per}}} \mathrm{d}\mathbf{i}}_{\text{Fisher}} - \underbrace{\rho_g \, \mathcal{J}_g^{C,r^{\mathrm{per}}} \left[(1+\theta)\mathbb{I} - \theta\rho_g L\right] \mathrm{d}\boldsymbol{\pi}}_{\text{Inflation}} \\ & - \underbrace{(1-\rho_g)\,\kappa_g \, \mathcal{J}_g^{C,r^{\mathrm{per}}} \left[\mathbb{I} - \theta\rho_g L\right] \mathrm{d}\chi_g}_{\text{Credibility}}.\end{aligned} \tag{37}$$

[6] In terms of notation, bold letters denote the stacked column vector of deviations at the different horizons, $\mathbf{X} = (X_0, X_1, \dots)'$.

Equation 37 shows that, with L the lag operator, the perceived-rate Jacobian decomposes into a pure Fisher channel, an inflation component $\rho_g \mathcal{J}_g^{C,r^{\text{per}}}$, and a credibility component $(1-\rho_g)\kappa_g \mathcal{J}_g^{C,r^{\text{per}}}$. The last one is the only channel that central-bank communication can influence, since its communication ν enters only through $m_{E,g}$.

To close the model, note that the cohort household blocks, the supply block (the Phillips curve), and the policy block jointly determine the aggregate paths $\mathbf{U} = (\mathbf{Y}, \boldsymbol{\pi}, \boldsymbol{\chi})'$ through the equilibrium conditions (32)–(A30). The model is solved in sequence space, taking output, inflation and credibility as unknowns, and imposing the asset-market clearing, the Phillips curve and the credibility law of motion conditions,

$$H\big(\mathbf{U}, \mathbf{Z}\big) = \mathbf{0}, \qquad \mathbf{Z} = (\varepsilon^{MP}, \varepsilon^{CP})', \tag{38}$$

where $\mathbf{Z}$ collects the exogenous monetary ε^{MP} and cost-push ε^{CP} shocks. Linearising around the steady state, the response of the aggregates to any shock is

$$\mathrm{d}\mathbf{U} = -H_{\mathbf{U}}^{-1} H_{\mathbf{Z}}\, \mathrm{d}\mathbf{Z}, \tag{39}$$

where $H_{\mathbf{U}}$ and $H_{\mathbf{Z}}$ are the corresponding Jacobians.

4.2.2. The Distorted-Beliefs Jacobian

This section makes explicit how to compute the household Jacobians under distorted beliefs. The main idea is to inject the belief distortion into the Euler equation and let the sequence-space solver differentiate the otherwise standard block. The implementation closely mirrors the behavioural-Jacobian representation of Guerreiro (2023) and Shabalina and Tzaawa-Krenzler (2024), in which a household block under non-full-information expectations can be written as the FIRE Jacobian operating on the believed path, plus a correction for forecast revisions.

For the paths of inflation and credibility, a convenient property of the diagnostic beliefs of Proposition 2 is $\mathrm{d}\mathbb{E}_g^{\theta}[\boldsymbol{\pi}] = \Lambda_{g,t}^{\pi} \mathrm{d}\boldsymbol{\pi} + \Lambda_{g,t}^{\chi} \mathrm{d}\boldsymbol{\chi}_g$, with $\Lambda_{g,t}^{\pi} = \rho_g[(1+\theta)\mathbb{I} - \theta\rho_g L]$ and $\Lambda_{g,t}^{\chi} = (1-\rho_g)\kappa_g[\mathbb{I} - \theta\rho_g L]$. Then, for $X \in \{\pi, \chi\}$, the behavioural Jacobian becomes:

$$\tilde{\mathcal{J}}_g^{C,X} = \mathcal{J}_g^{C,X} \Lambda_{g,0}^{X} + \sum_{t \geq 1} \mathcal{R}_{g,t}^{C,X} \left(\Lambda_{g,t}^{X} - \Lambda_{g,t-1}^{X}\right), \tag{40}$$

where $\mathcal{R}^{C,X}_{g,t}$ is the consumption response to a forecast revision at t. Intuitively, the first term is the FIRE Jacobian, while the second term contains all the forecast revisions that distorted beliefs trigger. Remarkably, forecast revisions disappear under Assumption 2 of Section 3. The reason is that ρ_g, θ and κ_g are independent of t. Therefore, the mapping from realised paths to expectations is never revised, and the belief matrices remain constant at $\Lambda^{\pi}_{g,t} = \Lambda^{\pi}_{g}$ and $\Lambda^{\chi}_{g,t} = \Lambda^{\chi}_{g}$, making the revision term in the behavioural Jacobian disappear as well. Lemma 3 formally states those results.

LEMMA 3 (The Distorted-Beliefs Jacobian). *Let $\mathcal{J}^{C,\pi}_g$ and $\mathcal{J}^{C,\chi}_g$ be the cohort-g consumption FIRE Jacobians with respect to the expected inflation path and the credibility path, and let $\widetilde{\mathcal{J}}^{C,\pi}_g$ and $\widetilde{\mathcal{J}}^{C,\chi}_g$ be the behavioural (distorted) Jacobians* (40). *Let $\Lambda^{\pi}_g = \rho_g[(1+\theta)\mathbb{I} - \theta\rho_g L]$ and $\Lambda^{\chi}_g = (1-\rho_g)\kappa_g[\mathbb{I} - \theta\rho_g L]$ be the operators of* (42) *stacked over horizons, with L the lag operator. Under Assumption 2 and Proposition 2:*

a. *the cohort-g belief is $\mathrm{d}\mathbb{E}^{\theta}_g[\boldsymbol{\pi}] = \Lambda^{\pi}_g \,\mathrm{d}\boldsymbol{\pi} + \Lambda^{\chi}_g \,\mathrm{d}\boldsymbol{\chi}_g$ at every date t, with $\Lambda^{\pi}_{g,t} = \Lambda^{\pi}_g$ and $\Lambda^{\chi}_{g,t} = \Lambda^{\chi}_g$ independent of t;*
b. *every forecast-revision term in* (40) *is zero, $\Lambda^{X}_{g,t} - \Lambda^{X}_{g,t-1} = 0$ for all $t \geq 1$ and $X \in \{\pi, \chi\}$, and the behavioural Jacobians collapse to*

$$\widetilde{\mathcal{J}}^{C,\pi}_g = \mathcal{J}^{C,\pi}_g \Lambda^{\pi}_g, \qquad \widetilde{\mathcal{J}}^{C,\chi}_g = \mathcal{J}^{C,\pi}_g \Lambda^{\chi}_g;$$

c. *the two paths reach consumption through the same perceived-rate Jacobian, so the FIRE Jacobians coincide,*

$$\mathcal{J}^{C,\pi}_g = \mathcal{J}^{C,\chi}_g = -\mathcal{J}^{C,r^{\mathrm{per}}}_g = -\mathcal{J}^{C,i}_g,$$

and Λ^{π}_g and Λ^{χ}_g are the two partial derivatives of the same subjective forecast.

PROOF. See Appendix A, subsection A.10. □

Lemma 3 generalizes the representative distorted consumption of the RA model to the HANK context, with the only difference that Λ^{π}_g and Λ^{χ}_g now multiply the entire intertemporal consumption Jacobian of cohort g. In contrast to Guerreiro (2023) and Shabalina and Tzaawa-Krenzler (2024), where the belief coefficients accumulate through learning and the revision terms are significant, the fixed ρ_g and $\bar{\pi}_g$ keep the mapping time-invariant. Therefore, even though this paper borrows the representation from those papers, the assumptions on belief formation place all learning dynamics in the credibility path χ_g.

Once diagnostic beliefs have been incorporated into the HANK architecture, Proposition 3 characterizes the two covariance statistics—one for beliefs, one for balance-sheet exposure—through which cohort heterogeneity reaches the aggregate consumption path.

PROPOSITION 3 (Aggregate Consumption in HANK). *Let* $\mathrm{d}\mathbf{C}_g$ *be the cohort-g consumption path generated by the Distorted-Beliefs Jacobians of Lemma 3 and* $\mathrm{d}\mathbf{C} = \sum_g \omega_g \mathrm{d}\mathbf{C}_g$. *Define the cohort's subjective-forecast path* $\mathrm{d}\mathcal{E}_g \equiv \Lambda_g^{\pi}\mathrm{d}\boldsymbol{\pi} + \Lambda_g^{\chi}\mathrm{d}\boldsymbol{\chi}_g$, *whose date-t element* $\mathrm{d}\mathcal{E}_{g,t} = \mathrm{d}\mathbb{E}_{g,t}^{\theta}[\pi_{t+1}]$ *is the change of cohort g's subjective inflation forecast,* $\mathrm{d}C_{g,t}^{X} \equiv \left[\mathcal{J}_g^{C,X}\,\mathrm{d}\mathbf{X}\right]_t$ *for* $X \in \{r, i, Y, \tau\}$, *with* $\overline{\mathrm{d}C^{X}}_t$ *its* ω*-weighted mean, and the two covariances*

$$\mathrm{CE}_t \equiv \mathrm{Cov}_{\omega}\big(\mathrm{MPC}_g, \mathrm{d}\mathcal{E}_{g,t}\big), \qquad \mathrm{CU} \equiv \mathrm{Cov}_{\omega}(\mathrm{MPC}_g, \mathrm{URE}_g).$$

a. *The aggregate response admits the exact decomposition*

$$\begin{aligned}\mathrm{d}C_t = &\underbrace{\left[\overline{\mathcal{J}^{C,\pi}}\,\overline{\mathrm{d}\mathcal{E}}\right]_t + \overline{\mathrm{d}C^{r}}_t}_{\textit{Representative Agent}} \\ &+ \underbrace{\Big[\sum_g \omega_g\big(\mathcal{J}_g^{C,\pi} - \overline{\mathcal{J}^{C,\pi}}\big)\big(\mathrm{d}\mathcal{E}_g - \overline{\mathrm{d}\mathcal{E}}\big)\Big]_t}_{\textit{Cross-cohort Heterogeneity}} \\ &+ \underbrace{\sum_{X\in\{i,Y,\tau\}} \overline{\mathrm{d}C^{X}}_t}_{\{i,Y,\tau\}\ \textit{Channel}}.\end{aligned} \tag{41}$$

b. *Because the inflation path is common across cohorts,* $\mathrm{CE}_t = \mathrm{CM}_t + \mathrm{CC}_t$, *with the diagnostic component* $\mathrm{CM}_t \equiv \mathrm{Cov}_{\omega}(\mathrm{MPC}_g, \Lambda_g^{\pi}\mathrm{d}\pi_t) = (1+\theta)\,\mathrm{Cov}_{\omega}(\mathrm{MPC}_g, \rho_g)\,\mathrm{d}\pi_t - \theta\,\mathrm{Cov}_{\omega}(\mathrm{MPC}_g, \rho_g^2)\,\mathrm{d}\pi_{t-1}$ *and the credibility component* $\mathrm{CC}_t \equiv \mathrm{Cov}_{\omega}(\mathrm{MPC}_g, \Lambda_g^{\chi}\mathrm{d}\chi_{g,t}) = -(1-\nu)\,\mathrm{Cov}_{\omega}\big(\mathrm{MPC}_g, (1-\rho_g)\Delta_g X_{g,t}\big)$, *with* $X_{g,t} \equiv \mathrm{d}\chi_{g,t} - \theta\rho_g\mathrm{d}\chi_{g,t-1}$. *Only* CC_t *depends on the communication intensity* ν.

c. *On impact, the forecast path carries no income effect, while the realised-rate channel satisfies exactly* $\sum_g \omega_g\,\mathcal{J}_{g,00}^{C,r} = \overline{\mathrm{MPC}}\,\overline{\mathrm{URE}} + \mathrm{CU}$. *If the cohort forecast Jacobians scale with the MPC profile, the cross-cohort heterogeneity block of* (41) *collapses into* $\frac{1}{\overline{\mathrm{MPC}}}\big[\overline{\mathcal{J}^{C,\pi}}\,\mathbf{CE}\big]_t$, *where* $\mathbf{CE} \equiv (\mathrm{CE}_s)_{s\geq 0}$.

Hence $(\mathrm{CE}_t, \mathrm{CU})$ *are sufficient statistics for determining the cohort heterogeneity on the aggregate consumption path.*

PROOF. See Appendix A, subsection A.11. □

Proposition 3 is useful for two reasons. First, it provides a general representation of the full consumption trajectory regardless of the shock the economy experiences. It decomposes it into a representative-agent term, a cross-cohort heterogeneity correction, and the $\{i, Y, \tau\}$ channels. Second, at every date t two covariance statistics govern heterogeneity, the Correlated-Expectations covariance (CE_t) and the Correlated-URE covariance (CU). While $CU = \mathrm{Cov}_\omega(MPC_g, URE_g)$ captures the impact income effect of the actual rate and is the standard redistribution channel of Auclert (2019), $CE_t = \mathrm{Cov}_\omega(MPC_g, d\mathcal{E}_{g,t})$ captures the covariance between the MPC and the movement of the subjective forecast itself. The parallel is with the correlated disagreement of Guerreiro (2023), with the difference that CE_t is a function of observables, $CE_t = CM_t + CC_t$. Hence, while CU is inherited from the standard HANK architecture, CE_t is inherent to the diagnostic-beliefs structure, and it contains the only margin—the credibility component CC_t—that central-bank communication can reach.

Additionally, Proposition 3 shows the limitations of central-bank communication on the composition of CE_t. By reshaping the cohort coefficients κ_g, communication moves the single component CC_t, while CM_t and CU are beyond its reach. Importantly, the sign of CM_t is structural. In HANK, the marginal propensity to consume falls with age and wealth, while the lived persistence ρ_g rises with age, as older cohorts experienced the persistent inflation of the 1970s and 1980s. Therefore, $\mathrm{Cov}_\omega(MPC_g, \rho_g) < 0$ and, with $\theta < 0$, $CM_t < 0$ whenever inflation is above its steady state. The sign of CC_t depends on the realised credibility paths. After a cost-push shock credibility falls by more for the cohorts with the larger experienced gap Δ_g, which are also the older, low-MPC cohorts, so $CC_t < 0$ as well.

Finally, Table 1 reports the calibration. Panel A collects the structural parameters of the three-equation block (23)–(25), set to the standard values of Auclert et al. (2024), together with the diagnostic parameter θ, imported from Carroll (2003). Since the DE operator imposes $\mathbb{E}^\theta_{g,t}[\pi_{t+1}] = \mathbb{E}_{g,t-1}[\pi_{t+1}] + (1+\theta)\big(\mathbb{E}_{g,t}[\pi_{t+1}] - \mathbb{E}_{g,t-1}[\pi_{t+1}]\big)$, the diagnostic expectation incorporates a fraction $1+\theta$ of the forecast revision on impact, which matches the share $\lambda = 0.27$ of households that update each quarter in Carroll (2003), so that $\theta = \lambda - 1 = -0.73$.

The learning parameters ρ_U and b complete the panel. Both are imported from Hommes and Lustenhouwer (2019): the memory of past forecast errors is $\rho_U = 0.5$ and the learning intensity is $b = 0.25$. Panel B builds the cohort moments from

UK data: the demographic weights ω_g are UK ONS mid-2019 shares, while the lifetime inflation persistence ρ_g and experienced mean $\bar{\pi}_g$ are constructed from the ONS quarterly-annualised CPI series (Appendix C.6.1), measured up to 2019Q4. It makes evident the heterogeneity across cohorts in experienced inflation. This experience determines how each cohort's anchor compares to the target, and it is ultimately the source of cohort heterogeneity when the central bank implements its communication strategy.

TABLE 1. **Calibration of the simple model.**

Panel A. Structural and belief parameters

Symbol	Value	Description	Source
β	0.99	Discount factor, quarterly	Auclert et al. (2024)
σ	1.00	Intertemporal elasticity of substitution	Auclert et al. (2024)
ϖ	0.03	New Keynesian Phillips curve slope	Auclert et al. (2024)
ϕ	1.50	Taylor-rule coefficient on inflation	Auclert et al. (2024)
ρ	0.50	AR(1) persistence of the shocks ε_t, ε_t^{CP}	Galí (2015)
π^*	2.00%	Inflation target	
θ	−0.73	Diagnostic parameter	Carroll (2003)
ρ_U	0.50	Memory of the discounted likelihood	Hommes and Lustenhouwer (2019)
b	0.25	Learning intensity	Hommes and Lustenhouwer (2019)
ν	0.00	Intensity of target-communication	
T	40	Horizon for the IRFs (quarters)	

Panel B. Cohort-level calibration

	15–24	25–34	35–44	45–54	55–64	65+	Pop. mean
Population weight, ω_g	0.148	0.163	0.153	0.168	0.146	0.222	1.000
Inflation persistence, ρ_g	0.113	0.106	0.035	0.179	0.559	0.554	0.274
Experienced inflation, $\bar{\pi}_g$ (%)	2.02	2.35	2.00	2.66	4.21	5.04	3.17

Notes: ρ_g and $\bar{\pi}_g$ are the AR(1) slope and the mean of the cohort's lifetime quarterly-annualised UK CPI inflation up to 2019Q4. The last column of Panel B reports population-weighted means.

5. Welfare and Central Bank Communication

This section makes the case that central bank communication has distributional effects when beliefs are distorted. In the RANK model with FIRE, every cohort shares the same forecast of future inflation and therefore picks the same consumption

path, so there are no redistributive concerns. However, under diagnostic beliefs this regularity no longer holds. Each cohort's experienced inflation and credibility shape its perceived real interest rate, and therefore, via its own Euler equation, its intertemporal consumption decisions. This heterogeneity in perceived rates leads to costly over- or under-reaction by cohorts relative to the average consumption path, with the direction and size of the deviation determined by each cohort's inflation experiences. Therefore, this divergence is the measure of redistribution used throughout the paper.

5.1. Consumption Wedges

Belief heterogeneity introduces cohort consumption divergence from the aggregate path, which causes significant welfare losses for those who deviate the most. I identify those off-path deviations in both models as follows. Define the cohort-g consumption wedge as the gap between the consumption path cohort g chooses under its *own* diagnostic belief (its own Euler equation) and the *equilibrium* aggregate path, which is the population average of the cohort paths: $\mathcal{W}_{g,t} \equiv dc_{g,t} - \overline{dc}_t$, with $\overline{dc}_t \equiv \sum_g \omega_g \, dc_{g,t}$. Importantly, the wedge is the same measure in the RANK and in the HANK model.

These wedges isolates the impact of belief heterogeneity on consumption, and constitutes an appropriate measure of welfare. Since the cohort loss is quadratic in its wedge, aggregating these off-optimal deviations delivers a measure of the welfare cost of belief heterogeneity and pins down which cohorts bear the largest share of that cost. In both models, once the equilibrium paths of inflation and credibility are known, the consumption wedges follow immediately. I first characterise the wedges in the RANK model with Proposition 4 , where only beliefs differ across cohorts, and then in the HANK model with Proposition 5, where balance sheets and income exposure differ as well.

PROPOSITION 4 (Consumption Wedges and Aggregate Welfare in RANK). *Consider any sequence of structural shocks that produces equilibrium paths* $\{\hat{\imath}_t, \hat{\pi}_t, \hat{\chi}_{g,t}\}_{t\geq 0}$ *of the RANK model in equations* (23)–(25). *Define* $dc_{g,t}$ *as the cohort-g consumption path that satisfies its own distorted Euler equation. Denote* $\overline{dc}_t \equiv \sum_g \omega_g \, dc_{g,t}$ *as the equilibrium aggregate path. The linearized inflation expectations follows* $\mathrm{d}\mathbb{E}^{\theta}_{g,t}[\hat{\pi}_{t+1}] = \Lambda^{\pi}_g \hat{\pi}_t + \Lambda^{\chi}_g \hat{\chi}_{g,t}$ *with*

$$\Lambda^{\pi}_g \hat{\pi}_t \;\equiv\; \rho_g\big[(1+\theta)\hat{\pi}_t - \theta\rho_g \hat{\pi}_{t-1}\big], \qquad \Lambda^{\chi}_g \hat{\chi}_{g,t} \;\equiv\; (1-\rho_g)\,\kappa_g\big[\hat{\chi}_{g,t} - \theta\rho_g \hat{\chi}_{g,t-1}\big], \quad (42)$$

and let $\bar{\Lambda}^{\pi} \equiv \sum_g \omega_g \Lambda_g^{\pi}$ *and* $\overline{\Lambda^{\chi}\hat{\chi}}_t \equiv \sum_g \omega_g \Lambda_g^{\chi}\hat{\chi}_{g,t}$ *denote their population averages.*

Let $A_g \equiv -U_g''(c_{g,t}) > 0$ *denote cohort g's coefficient of absolute risk aversion at its subjective optimum. Then, the* ***cohort-g consumption wedge****,* $\mathcal{W}_{g,t} \equiv dc_{g,t} - \overline{dc}_t$ *has the close-form expression:*

$$\mathcal{W}_{g,t} \;=\; \sigma \sum_{s\geq 0}\Big[\underbrace{\big(\Lambda_g^{\pi} - \bar{\Lambda}^{\pi}\big)\hat{\pi}_{t+s}}_{\textit{Diagnostic}} \;+\; \underbrace{\Lambda_g^{\chi}\hat{\chi}_{g,t+s} - \overline{\Lambda^{\chi}\hat{\chi}}_{t+s}}_{\textit{Credibility}}\Big]. \tag{43}$$

Furthermore, ***cohort-g welfare loss*** *depends non-linearly on* $\mathcal{W}_{g,t}$*,*

$$L_{g,t} \;\equiv \tfrac{1}{2}\big(\mathcal{W}_{g,t}\big)^2, \tag{44}$$

while the ***aggregate welfare loss*** *is independent of the aggregate consumption path and equals the cross-cohort dispersion of the cohort paths,*

$$\mathcal{L}_t \;=\; \tfrac{1}{2}\,\mathrm{Var}_{\omega}\big(dc_{g,t}\big). \tag{45}$$

PROOF. See Appendix A, subsection A.6. □

In the RANK model, Proposition 4 clarifies that, regardless of the type of shock hitting the economy, both the cohort-g and the aggregate welfare losses depend only on the equilibrium paths $(\hat{\pi}_t, \hat{\chi}_{g,t})$. However, since credibility responds endogenously to inflation, an inflationary shock changes central-bank credibility by moving $\hat{\pi}_t$ directly and then, it spreads out through the credibility channel $\hat{\chi}_{g,t}$. A qualitative implication of Proposition 4 is that inflationary shocks redistribute initially via *how strongly each cohort extrapolates* the shock into expected future inflation, captured by Λ_g^{π}, which is ultimately disciplined by its lifetime inflation persistence ρ_g. Additionally, there is another fundamental margin that governs *toward which anchor* the forecast inclines, captured by $\kappa_g = -(1-\nu)\Delta_g$ and activated by $\hat{\chi}_{g,t}$. Thus, an inflationary shock generates an additional effect on consumption wedges by eroding credibility since cohorts that have consistently experienced inflation above the target expect more inflation and therefore bear larger consumption wedges.

I turn next to calculating the consumption wedges for the HANK model in Proposition 5 by explicitly showing the effect of the URE channel. In this case, the exposure terms enter alongside the distorted beliefs component, so the wedge no longer depends only on the equilibrium paths of inflation and credibility, but also on each cohort's balance sheet and income.

PROPOSITION 5 ((Consumption Wedges and Aggregate Welfare in HANK). *Let* $\mathrm{d}\mathbf{C}_g$ *be the cohort-g consumption path generated by* (36)*, with the perceived-rate path in* (37)*. Define the cohort-g per-capita consumption path as* $\mathrm{d}c_{g,t} \equiv \mathrm{d}C_{g,t}/C_g$*, the Jacobians* $\mathcal{J}_g^{c,X} \equiv \mathcal{J}_g^{C,X}/C_g$*, and* $\mathcal{J}_g^{c,\pi} \equiv -\mathcal{J}_g^{c,r^{\text{per}}}$*, the consumption response to the cohort's subjective inflation forecast. Let* Λ_g^{π} *and* Λ_g^{χ} *be defined by* (42)*. Then, the cohort consumption wedge* $\mathcal{W}_{g,t}^{H} \equiv \mathrm{d}c_{g,t} - \overline{\mathrm{d}c}_t$ *is:*

$$\begin{aligned}\mathcal{W}_{g,t}^{H} = & \underbrace{\left[\left(\mathcal{J}_g^{c,\pi}\Lambda_g^{\pi} - \overline{\mathcal{J}^{c,\pi}\Lambda^{\pi}}\right)\mathrm{d}\boldsymbol{\pi} + \mathcal{J}_g^{c,\pi}\Lambda_g^{\chi}\mathrm{d}\chi_g - \overline{\mathcal{J}^{c,\pi}\Lambda^{\chi}\mathrm{d}\chi}\right]_t}_{\textit{Distorted-Beliefs Dispersion}} \\ & + \underbrace{\left[\left(\mathcal{J}_g^{c,r} - \overline{\mathcal{J}^{c,r}}\right)\mathrm{d}\mathbf{r}\right]_t}_{\textit{Balance-Sheet (URE) Dispersion}} \\ & + \underbrace{\sum_{X\in\{i,Y,\tau\}}\left[\left(\mathcal{J}_g^{c,X} - \overline{\mathcal{J}^{c,X}}\right)\mathrm{d}\mathbf{X}\right]_t}_{\{i,Y,\tau\}\textit{ Channel}}.\end{aligned} \tag{46}$$

with $\sum_g \omega_g \mathcal{W}_{g,t}^{H} = 0$*. Moreover, the cohort welfare loss is given by* $L_{g,t} = \frac{1}{2}(\mathcal{W}_{g,t}^{H})^2$ *and the aggregate loss is* $\mathcal{L}_t = \frac{1}{2}\,\mathrm{Var}_{\omega}(\mathrm{d}c_{g,t})$*.*

PROOF. See Appendix A, subsection A.12. □

Proposition 5 separates each wedge into the distorted beliefs component as in the RANK case, and two additional exposure terms, namely the balance-sheet (URE) term on the realised real rate and the $\{i, Y, \tau\}$ components on the nominal rate, income and taxes. Therefore, the HANK model adds two things to the RANK wedge. First, the belief element captures via $\mathcal{J}_g^{c,\pi}$, the situation that the same forecast deviation changes consumption by more in a high-MPC cohort than in a low-MPC one. Second, the exposure terms, which are zero in the RANK model, capture how cohort's differences in their asset positions and in their income affects wedges. Particularly, younger cohorts with URE below the mean experience negative consumption wedges since they hold the smallest asset positions and have the highest MPC, so they gain the least from the rise in the real rate, which makes them under-consume relative to the average regardless of their distorted beliefs.

FIGURE 5. Cohort Consumption Wedges After a +1pp Cost-Push

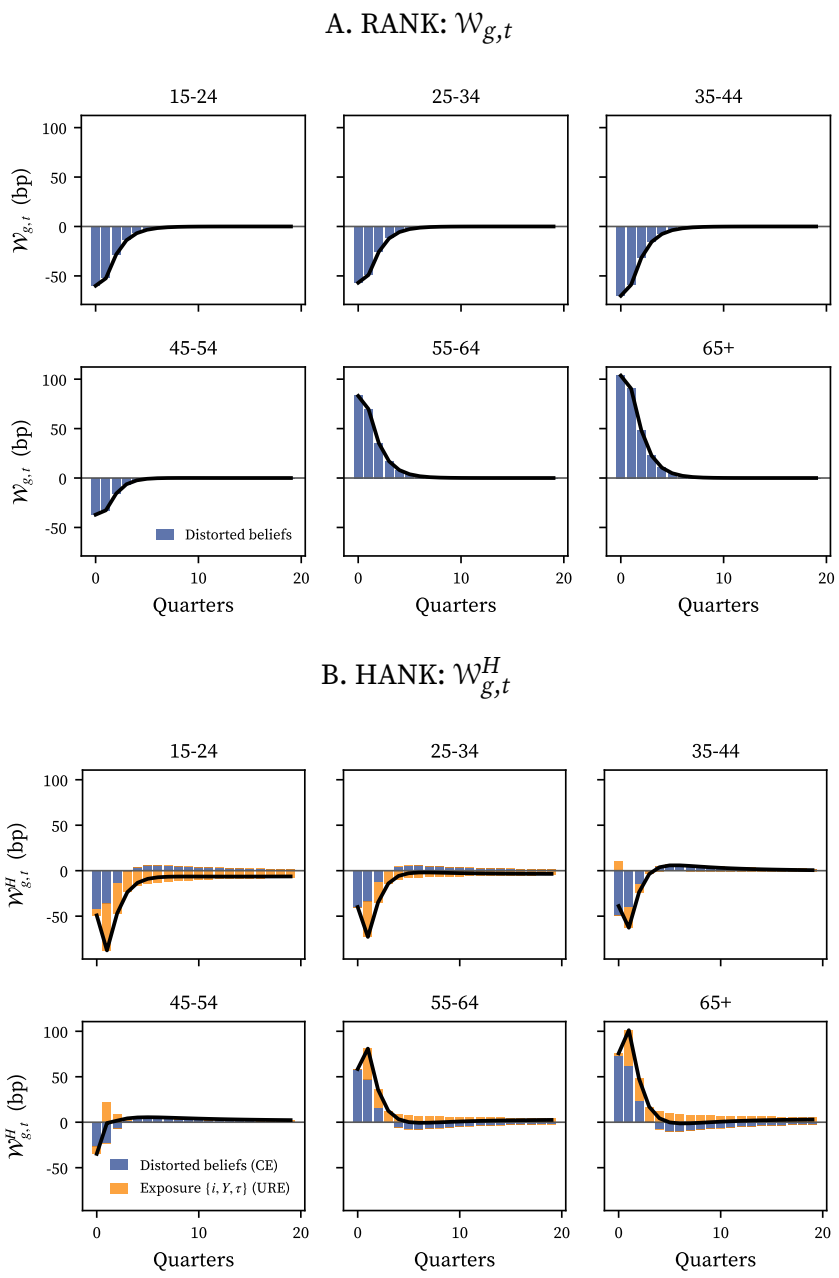


Figure 5 shows the cohort wedges after a cost-push for both models. In the RANK model, panel (a), consumption wedges are entirely composed of distorted beliefs because the only reason its consumption departs from the average is that it forecasts inflation differently. In the HANK model, panel (b), the figure explicitly shows the role of the exposure channel in determining consumption wedges. Particularly, this channel is negative and largest for the younger cohorts. They

are strongly affected by the income contraction after the inflationary shock since they depend mostly on that income since they asset position is low and they hold the highest MPC. Therefore, they are forced to cut consumption by more than the average, leaving them with the highest welfare losses in general.

5.2. Communication Design

This section elaborates on the question of which communication intensity the central bank should choose to limit the suboptimal swings in consumption due to inflation experiences and credibility. Additionally, this section shows how the incidence of each cohort in the aggregate loss changes from the RANK to the HANK model due to the URE channel. A second result presented here is the conditional complementarity between communication and traditional monetary policy, that is, a policy that after an inflationary shock, calls for the interest rate to rise by the magnitude of the inflation response coefficient of the Taylor rule. Finally, the section elaborates on the question about the loss incurred by a late implementation of target communication, showing that, for both models, early communication drastically reduce the loss incurred by its counterpart delayed reaction.

5.2.1. Optimnal Target Communication

Intuitively, the distorted beliefs mechanism affects more intensively the cohorts with a high inflationary memory and a credibility exposure above the mean, because when a cost-push shock hits the economy, their expectations rise by more than the average, pulling them back to their own inflation history, and therefore they bring consumption forward suboptimally. Proposition 6 tackles the optimal target-based communication that the CB may implement to restrict these consumption swings.

PROPOSITION 6 (Optimal Target Communication). *Consider a cost-push shock with realised, already observed, equilibrium paths $\{\hat{\pi}_t, \hat{\chi}_{g,t}\}_{t\geq 0}$ of inflation and cohort credibility.*

a. RANK Model. *Let the ex-post credibility contribution to the cohort-g wedge of Proposition 4 be*

$$\mathcal{W}_{g,t}^{ex\text{-}post,\chi}(\nu) \equiv \sigma \sum_{s\geq 0}\Big[\Lambda_g^{\chi}\hat{\chi}_{g,t+s} - \overline{\Lambda^{\chi}\hat{\chi}}_{t+s}\Big],$$

with losses $L_{g,t}^{ex\text{-}post,\chi}(\nu) = \frac{1}{2}\left(\mathcal{W}_{g,t}^{ex\text{-}post,\chi}\right)^2$ *and* $\mathcal{L}_t^{ex\text{-}post,\chi}(\nu) = \sum_g \omega_g L_{g,t}^{ex\text{-}post,\chi}(\nu)$. *Then the intensity cohort g would choose and the aggregate optimum coincide when* $\nu = 1$,

$$\nu_g^\star \equiv \arg\min_{\nu\in\mathbb{R}} L_{g,t}^{ex\text{-}post,\chi}(\nu) = 1, \qquad \nu_{\text{agg}}^\star \equiv \arg\min_{\nu\in\mathbb{R}} \mathcal{L}_t^{ex\text{-}post,\chi}(\nu) = 1.$$

b. HANK Model. *Let the ex-post credibility contribution to the cohort-g wedge of Proposition 5 be*

$$\mathcal{W}_{g,t}^{H,\chi}(\nu) \equiv \left[\mathcal{J}_g^{c,\pi}\Lambda_g^\chi \mathrm{d}\chi_g - \overline{\mathcal{J}^{c,\pi}\Lambda^\chi \mathrm{d}\chi}\right]_t,$$

with losses $L_{g,t}^{H,\chi}(\nu) = \frac{1}{2}\left(\mathcal{W}_{g,t}^{H,\chi}\right)^2$ *and* $\mathcal{L}_t^{H,\chi}(\nu) = \sum_g \omega_g L_{g,t}^{H,\chi}(\nu)$. *Then*

$$\nu_g^\star \equiv \arg\min_{\nu\in\mathbb{R}} L_{g,t}^{H,\chi}(\nu) = 1, \qquad \nu_{\text{agg}}^\star \equiv \arg\min_{\nu\in\mathbb{R}} \mathcal{L}_t^{H,\chi}(\nu) = 1,$$

as in the RANK model.

PROOF. See Appendix A, subsection A.7. □

Proposition 6 makes precise what optimal communication can achieve under an inflationary shock. Consistently across models, Propositions 4 and 5 show that the credibility term of every cohort's wedge is proportional to $1 - \nu$, so the central bank removes it entirely by uniformly communicating the target at its maximal intensity. Hence, there is no trade-off across cohorts in the choice of a common ν. Remarkably, in the HANK model the same credibility deviation moves consumption by more in a high-MPC cohort than in a low-MPC one, so heterogeneous MPCs change the size of the credibility loss and which cohorts carry the highest burden, but the central bank communication intensity that removes it remains unaffected.

However, the central bank faces limitations. First, it cannot reach the diagnostic channel, since ν only affects κ_g but never ρ_g or θ. This implies that the dispersion of Λ_g^π is unchanged by how the central bank decides to communicate the target, causing a positive welfare loss coming from the diagnostic term of the consumption wedge. Second, the central bank also cannot affect the balance-sheet and income exposures of the HANK wedge, which are fixed by the cohorts' asset positions and MPCs. Therefore, CB communication only governs the credibility channel of the redistribution, while the diagnostic component is pinned down by each cohort's inflation experiences ρ_g and is out of its reach.

FIGURE 6. Central Bank Communication and Welfare

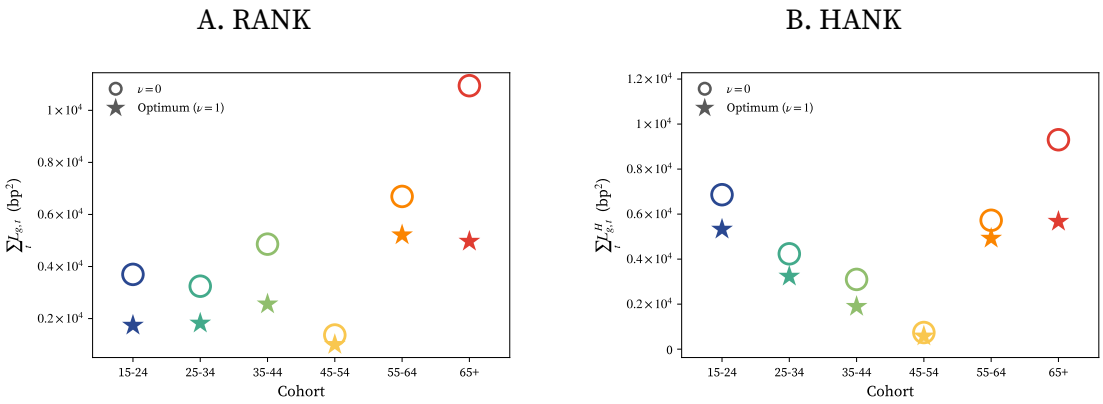


Figure 6 reports the cumulative loss of each cohort under the cost-push shock at $\nu = 0$ and at the aggregate optimum, which coincides with maximal intensity in both models, as Proposition 6 states. In the RANK model, panel A, maximal intensity lowers the loss of every cohort, which makes it a Pareto improvement over non-active target communication, and mainly the oldest cohort, which experiences the largest credibility deviation from the mean, gets the largest gain from $\nu = 1$. In the HANK model, panel B, the URE channel adds a second source of incidence that mainly affects the young cohorts. Since they hold the smallest asset positions and have the highest MPC, they are heavily affected by the rise in the realised real rate after the inflationary shock. This causes a drastic fall in their consumption and now their loss is comparable to that of the oldest cohort, a situation absent in the RANK model. Therefore, the burden is no longer concentrated at the top of the age distribution due to the URE channel in the HANK model.

Finally, Figure 7 shows how the share of the aggregate loss carried by each cohort, Ω_g,

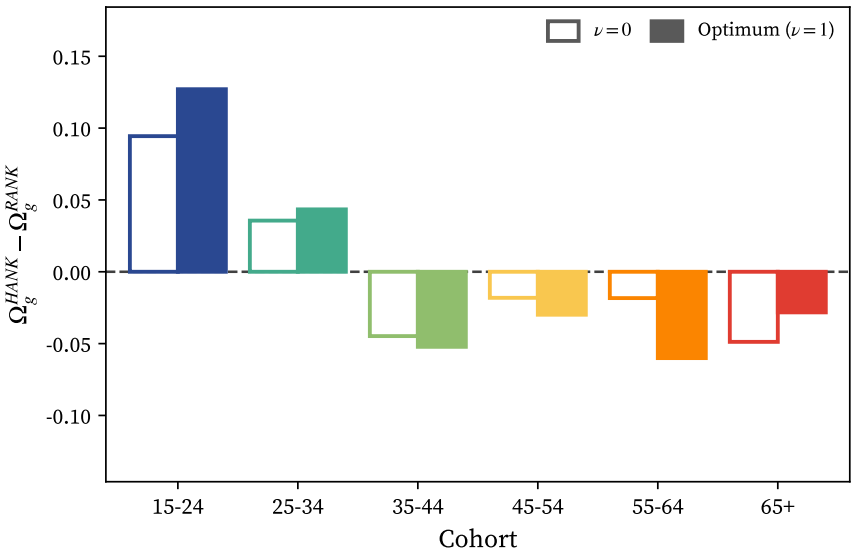
$$\Omega_g \equiv \frac{\omega_g \sum_t L_{g,t}}{\sum_h \omega_h \sum_t L_{h,t}}, \qquad \sum_g \Omega_g = 1. \tag{47}$$

changes once heterogeneous MPCs and the URE channel are active, as in the HANK model. To show this, I define the *communication incidence* as the difference between Ω_g^{HANK}, the shares in HANK, and Ω_g^{RANK}, the shares in the RANK model. Incidence measures the shift of the burden from those who remember differently due to their distorted beliefs (RANK model), to those who spend differently (HANK model).

In the RANK model, the shares of aggregate loss are pure distorted beliefs,

since the share of each cohort only reflects how far its distorted beliefs are from the average. Then, older cohorts carry most of the loss because their inflation history is the most persistent and the farthest from the target. In HANK the shares move towards a spending motive. A deviation in expectations through the beliefs channel becomes a loss only if the household adjusts consumption, which ultimately depends on its MPC and on its URE. In this case, young households are the most affected, given their high MPC and their small asset positions. Finally, communication consistently changes the size of the shift but not its direction.

FIGURE 7. Cohort Incidence of the Cost-Push Shock (+1pp).



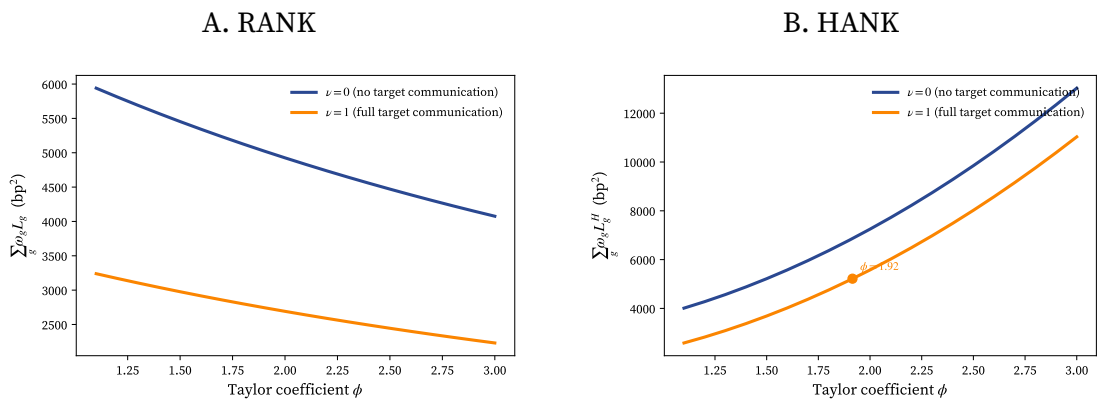
5.2.2. Communication, Monetary Policy and Timing

Once full target communication is proved optimal, its implementation raises two essential questions: how target communication interacts with the traditional interest-rate adjustments, as asked by Goy et al. (2022) for forward guidance under imperfect credibility and empirically by Coibion et al. (2022), and, once the economy is hit by an inflationary shock, what the associated costs of delayed communication are, in the timing sense of Bomfim and Rudebusch (2000) for disinflation under imperfect credibility.

Figure 8 compares CB communication with the inflation coefficient of the Taylor rule, ϕ. In the RANK model, panel A, a more aggressive rule monotonically reduces the loss. However, without target communication no ϕ is able to deliver

a loss as low as that of full communication, so CB communication cannot be substituted by traditional monetary policy: the two policies are complements, and the lowest loss is attained by communicating the target intensively together with an aggressive rate response. On the other hand, in the HANK model, panel B, the curves reverse because a stronger interest rate response affects cohorts through the exposure channel. In contrast, in this case communication and interest rate adjustments are substitutes, since different combinations of communication and aggressiveness in the interest rate deliver the same welfare loss. In particular, when $\nu = 1$ the central bank can afford a tighter rule, while without target communication the same welfare loss calls for a more moderate rate response. Hence, the lowest loss combines intense target communication with a passive monetary policy to limits the distributional channel that URE imposes.

FIGURE 8. Target Communication and the Inflation Coefficient in the Taylor Rule



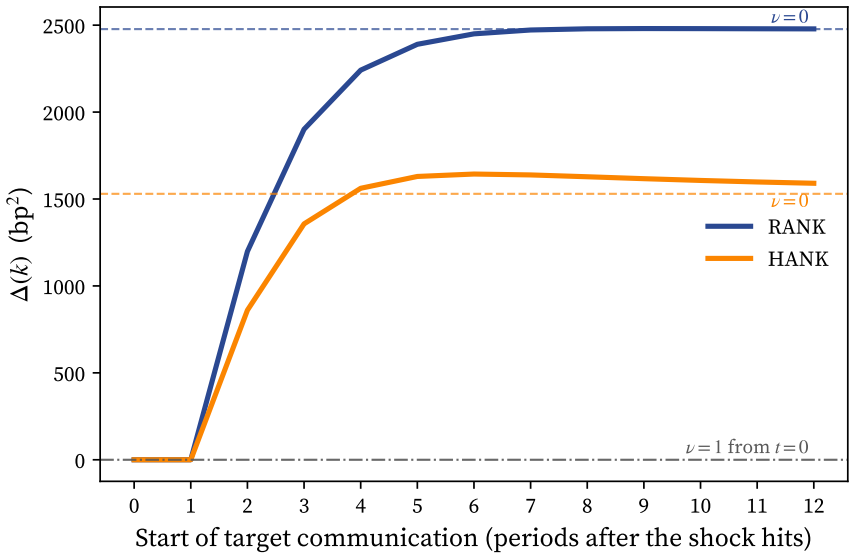
Finally, Figure 9 addresses the timing question by reporting the aggregate loss when complete target communication ($\nu = 1$) starts k periods after the shock, against the $\nu = 0$ reference case. Namely, it measures the incurred aggregate loss of starting late to fully communicate the target. Formally, let $\nu^{(k)}$ be the intensity path that adopts full target communication k periods after the shock hits the economy. The cost of the delayed communication switch is:

$$\Delta(k) \equiv \mathcal{L}(k) - \mathcal{L}(0), \qquad \mathcal{L}(k) \equiv \sum_g \omega_g \sum_{t \geq 0} L_{g,t}\big(\nu^{(k)}\big). \tag{48}$$

where $L_{g,t}(\nu^{(k)})$ is the cohort loss of (44) on the equilibrium path under $\nu^{(k)}$, $\mathcal{L}(0)$ is the loss under full communication from the impact and $\mathcal{L}(k)$ the loss if target communication starts only at period k. For both models, reacting sooner

rather than later delivers a lower welfare loss. This result is intuitive, given that credibility reacts to lagged inflation and the credibility channel of the wedge opens one period after the shock. Therefore, the target has to be communicated when the shock hits, not once its effects on expectations are visible via the credibility erosion.

FIGURE 9. Timing of Target Communication



6. Conclusion

Central banks respond to inflationary episodes by constantly repeating that they will do everything in their power to bring inflation back to the target. While this message seems to reach the most sophisticated individuals, households apparently do not notice these announcements when forming their expectations. This paper builds a model of inflation expectations with endogenous credibility that is consistent with the under-reaction of households by incorporating DE as a fundamental part of how households process news, and then applies this mechanism to the New Keynesian model, both in its simple version and in its heterogeneous-agent version.

Three lessons emerge from this paper. The first is related to who pays more in terms of welfare after an inflationary shock. When only beliefs differ, the cost falls on the cohorts whose memories diverge the most from the target, namely the old cohorts. Once wealth and income heterogeneity are incorporated, the real

rate revalues each cohort's position through the URE channel, and the cost moves towards the younger cohorts, which adjust consumption the most due to their high MPCs and small asset positions. Therefore, while distorted beliefs decide who misperceives inflation, the balance-sheet channel decides who pays for it. The second lesson is that intensively communicating the target is always optimal, regardless of whether the exposure channel is considered or not. The third lesson is about the connection between target-based communication and traditional monetary policy. With a representative agent, a more aggressive interest rate rule with intense target communication reinforce each other. However, with heterogeneous agents, an aggressive monetary response causes redistribution to the young cohorts, therefore communication and a higher response to inflation in the Taylor rule turn to become substitutes.

This paper leaves some open questions. The communication intensity is a choice of the CB, but its reception by households is taken as given. Studying theoretically how the central bank can reach households with its communication is a research area that would complement this paper. Another pending area comes from relaxing the assumption that inflation experiences are fixed over time. Allowing recent inflation to be incorporated into experienced inflation presents a methodological challenge, but it would evidently add realism to this analysis. Finally, extending the mechanism to the interaction between target communication and forward guidance on the policy rate would offer a wider spectrum of communication policies that the CB can use under distorted expectations.

Online Appendix

The Online Appendix is available **here.**

References

Angeletos, G.-M. and Sastry, K. A. (2021). Managing expectations: Instruments versus targets. *Quarterly Journal of Economics*, 136(4):2467–2532.

Auclert, A. (2019). Monetary policy and the redistribution channel. *American Economic Review*, 109(6):2333–67.

Auclert, A., Bardóczy, B., Rognlie, M., and Straub, L. (2021). Using the sequence-space jacobian to solve and estimate heterogeneous-agent models. *Econometrica*, 89(5):2375–2408.

Auclert, A., Rognlie, M., and Straub, L. (2024). The intertemporal keynesian cross. *Journal of Political Economy*, 132(12):4068–4121.

Bank of England (2022). Monetary policy report, november 2022. Bank of England, 3 November 2022.

Bianchi, F., Ilut, C., and Saijo, H. (2024). Diagnostic business cycles. *Review of Economic Studies*, 91(1):129–162.

Binder, C. (2017). Fed speak on main street: Central bank communication and household expectations. *Journal of Macroeconomics*, 52:238–251.

Blinder, A. S. (2000). Central-bank credibility: Why do we care? how do we build it? *American Economic Review*, 90(5):1421–1431.

Bomfim, A. N. and Rudebusch, G. D. (2000). Opportunistic and deliberate disinflation under imperfect credibility. *Journal of Money, Credit and Banking*, 32(4):707–721.

Bordalo, P., Gennaioli, N., Ma, Y., and Shleifer, A. (2020). Overreaction in macroeconomic expectations. *American Economic Review*, 110(9):2748–2782.

Bordalo, P., Gennaioli, N., and Shleifer, A. (2018). Diagnostic expectations and credit cycles. *The Journal of Finance*, 73(1):199–227.

Bordalo, P., Gennaioli, N., and Shleifer, A. (2022). Overreaction and diagnostic expectations in macroeconomics. *Journal of Economic Perspectives*, 36(3):223–244.

Bordo, M. D. and Siklos, P. L. (2015). Central bank credibility before and after the crisis. NBER Working Paper 21710, National Bureau of Economic Research, Cambridge, MA.

Born, B., Enders, Z., Menkhoff, M., Müller, G. J., and Niemann, K. (2025). Firm expectations and news: Micro v macro. Working paper, University of Bonn, Heidelberg University, University of Copenhagen and University of Tübingen.

Carroll, C. D. (2003). Macroeconomic expectations of households and professional forecasters. *The Quarterly Journal of Economics*, 118(1):269–298.

Christelis, D., Georgarakos, D., Jappelli, T., and van Rooij, M. (2020). Trust in the central bank and inflation expectations. *International Journal of Central Banking*, 16(6):1–37.

Coibion, O. and Gorodnichenko, Y. (2015). Information rigidity and the expectations

formation process: A simple framework and new facts. *American Economic Review*, 105(8):2644–2678.

Coibion, O., Gorodnichenko, Y., and Weber, M. (2022). Monetary policy communications and their effects on household inflation expectations. *Journal of Political Economy*, 130(6):1537–1584.

D'Acunto, F., Hoang, D., Paloviita, M., and Weber, M. (2020). Effective policy communication: Targets versus instruments. Becker Friedman Institute Working Paper 2020-148, University of Chicago.

Ehrmann, M., Georgarakos, D., and Kenny, G. (2025). Credibility gains from central bank communication with the public. *European Economic Review*, 177:105069.

Galí, J. (2015). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications*. Princeton University Press, 2 edition.

Goy, G., Hommes, C., and Mavromatis, K. (2022). Forward guidance and the role of central bank credibility under heterogeneous beliefs. *Journal of Economic Behavior and Organization*, 200:1240–1274.

Guerreiro, J. (2023). Belief disagreement and business cycles. Working paper, Northwestern University.

Haldane, A. and McMahon, M. (2018). Central bank communications and the general public. *AEA Papers and Proceedings*, 108:578–583.

Hommes, C. and Lustenhouwer, J. (2019). Inflation targeting and liquidity traps under endogenous credibility. *Journal of Monetary Economics*, 107:48–62.

Koop, G. and Korobilis, D. (2012). Forecasting inflation using dynamic model averaging. *International Economic Review*, 53(3):867–886.

Kumar, S., Afrouzi, H., Coibion, O., and Gorodnichenko, Y. (2015). Inflation targeting does not anchor inflation expectations: Evidence from firms in new zealand. *Brookings Papers on Economic Activity*, 2015(2):151–225.

Lagarde, C. (2022). Monetary policy in a high inflation environment: Commitment and clarity. Lecture organised by Eesti Pank, Tallinn, 4 November 2022.

Lamla, M. J. and Vinogradov, D. V. (2019). Central bank announcements: Big news for little people? *Journal of Monetary Economics*, 108:21–38.

L'Huillier, J.-P., Singh, S. R., and Yoo, D. (2024). Incorporating diagnostic expectations into the new keynesian framework. *Review of Economic Studies*, 91(5):3013–3046.

Malmendier, U. and Nagel, S. (2016). Learning from inflation experiences. *The Quarterly Journal of Economics*, 131(1):53–87.

McMahon, M. and Rholes, R. (2025). Building central bank credibility: The role of forecast performance. Working paper.

Nakamura, E. and Steinsson, J. (2018). High-frequency identification of monetary non-

neutrality: The information effect. *The Quarterly Journal of Economics*, 133(3):1283–1330.

Powell, J. H. (2022). Monetary policy and price stability. Speech at the Jackson Hole Economic Policy Symposium, Federal Reserve Bank of Kansas City, 26 August 2022.

Powell, J. H. (2023). Inflation: Progress and the path ahead. Speech at the Jackson Hole Economic Policy Symposium, Federal Reserve Bank of Kansas City, 25 August 2023.

Romer, C. D. and Romer, D. H. (2000). Federal reserve information and the behavior of interest rates. *American Economic Review*, 90(3):429–457.

Salle, I., Gorodnichenko, Y., and Coibion, O. (2023). Lifetime memories of inflation: Evidence from surveys and the lab. NBER Working Paper 31996, National Bureau of Economic Research, Cambridge, MA.

Shabalina, E. and Tzaawa-Krenzler, M. (2024). Heterogeneous attention to inflation and monetary policy. Rba workshop paper, Reserve Bank of Australia.

Young, E. R. (2010). Solving the incomplete markets model with aggregate uncertainty using the krusell-smith algorithm and non-stochastic simulations. *Journal of Economic Dynamics and Control*, 34(1):36–41.